

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Lab Resource Utilization Platform

Document Version: 1.0

Document Owner: Product Management Team

Project Name: Lab Resource Utilization Platform

Project Type: Enterprise Full Stack Web Application

Organization: Infosys Springboard Internship Project

Prepared By: Product Team

Technology Stack:

- Frontend: React.js
- Backend: Spring Boot (Java)
- Database: PostgreSQL
- Authentication: Spring Security + JWT + OAuth2
- Cache: Redis
- Cloud Storage: AWS S3 / Cloudinary
- Deployment: Docker + AWS
- Messaging: Kafka / RabbitMQ (Future Scope)

Project Duration: 8 Weeks

Document Status: Final Draft

Version History

Version	Date	Author	Description
0.1	Initial Draft	Product Team	Requirement Gathering

0.5	Internal Review	Development Team	Added Functional Modules
1.0	Final	Product Manager	Approved PRD

Table of Contents

1. Executive Summary
 2. Problem Statement
 3. Vision
 4. Business Objectives
 5. Project Scope
 6. Stakeholders
 7. Assumptions
 8. Constraints
 9. Success Criteria
 10. Glossary
-

1. Executive Summary

The **Lab Resource Utilization Platform (LRUP)** is a centralized enterprise-grade laboratory resource management system designed to improve the utilization, accessibility, and operational efficiency of laboratory equipment across universities, research institutions, and educational organizations.

Modern institutions invest substantial capital in advanced laboratory equipment such as electron microscopes, CNC machines, spectroscopy instruments, 3D printers, biomedical devices, and GPU servers. Despite these investments, much of the equipment remains underutilized due to fragmented management processes, manual booking systems, poor visibility into equipment availability, and the absence of centralized scheduling.

The Lab Resource Utilization Platform addresses these challenges by providing a unified web-based solution that enables institutions to:

- Maintain a centralized equipment inventory.
- Monitor equipment availability in real time.
- Schedule and manage bookings efficiently.
- Share resources between institutions.
- Track maintenance and calibration schedules.
- Generate analytical reports and utilization insights.
- Reduce equipment downtime and idle resources.
- Support data-driven procurement decisions.

The platform will be developed using a modern full-stack architecture with **React.js** as the frontend, **Spring Boot** as the backend, and **PostgreSQL** as the primary database. Secure access will be implemented using **Spring Security**, **JWT**, and **OAuth2**.

The application is intended to be scalable, modular, secure, and cloud-ready.

2. Problem Statement

Research laboratories and higher education institutions spend significant budgets on acquiring sophisticated laboratory equipment. However, the management of these resources is often decentralized and inefficient.

The current process generally involves:

- Manual registers or spreadsheets.
- Email or phone-based booking requests.
- Limited visibility into equipment availability.
- Inconsistent maintenance tracking.
- Lack of standardized workflows.
- No centralized utilization analytics.

These operational inefficiencies result in:

Business Problems

- Low utilization of expensive equipment.
- Duplicate procurement of resources.
- Increased operational expenses.
- Frequent scheduling conflicts.
- Delayed research activities.
- Poor maintenance compliance.
- Lack of transparency across departments.

Technical Problems

- No centralized inventory database.
- Absence of role-based access control.
- Lack of real-time booking information.
- No automated notification system.
- No predictive maintenance tracking.
- Limited reporting capabilities.

Without an integrated platform, institutions are unable to maximize the return on investment (ROI) of their laboratory infrastructure.

3. Vision Statement

To develop a scalable, secure, and intelligent laboratory resource management platform that enables institutions to optimize laboratory equipment utilization, simplify resource scheduling, facilitate inter-institution collaboration, and provide actionable insights through centralized analytics.

The platform should become the single source of truth for laboratory resource management.

4. Mission Statement

Our mission is to transform traditional laboratory management by delivering a modern enterprise platform that:

- Simplifies laboratory operations.
 - Improves equipment accessibility.
 - Enables efficient scheduling.
 - Promotes collaboration between institutions.
 - Supports research productivity.
 - Reduces operational costs.
 - Improves asset lifecycle management.
-

5. Business Objectives

The primary business objectives of this platform are:

BO-01

Increase laboratory equipment utilization by at least **80%**.

BO-02

Reduce idle equipment time by **30%**.

BO-03

Provide centralized visibility into all laboratory assets.

BO-04

Automate laboratory booking workflows.

BO-05

Enable equipment sharing between multiple institutions.

BO-06

Reduce duplicate procurement costs.

BO-07

Improve maintenance planning and equipment lifecycle management.

BO-08

Generate accurate utilization reports for management.

BO-09

Improve researcher productivity through simplified equipment access.

BO-10

Support strategic decision-making using analytics.

6. Technical Objectives

The system shall:

- Support secure authentication.
 - Provide role-based authorization.
 - Expose RESTful APIs.
 - Be scalable for multiple institutions.
 - Support cloud deployment.
 - Maintain modular architecture.
 - Support future microservice migration.
 - Maintain audit logs.
 - Support notification services.
 - Provide responsive web interfaces.
-

7. Project Scope

In Scope

The following functionalities are included in Version 1.0:

Authentication

- User Registration
 - Login
 - JWT Authentication
 - OAuth Login
 - Password Reset
 - Role Management
-

Institution Management

- Institution Registration
 - Department Management
 - Laboratory Management
-

Equipment Inventory

- Equipment Registration
 - Equipment Categories
 - QR Code Support
 - Equipment Documentation
 - Equipment Availability
-

Booking

- Equipment Search
 - Calendar Scheduling
 - Booking Requests
 - Booking Approval
 - Waitlist
 - Booking History
-

Utilization Monitoring

- Equipment Utilization
 - Daily Reports
 - Monthly Reports
 - Department Statistics
 - Utilization Heatmaps
-

Maintenance

- Maintenance Scheduling
 - Technician Assignment
 - Calibration Tracking
 - Service History
-

Resource Sharing

- Institution Discovery
 - Equipment Sharing
 - Access Approval
 - Shared Booking
-

Notifications

- Email Notifications
 - SMS Alerts
 - Push Notifications
-

Analytics

- Dashboard
 - Equipment ROI
 - Department Reports
 - Utilization Reports
-

Billing

- Usage Charges
- Cost Allocation
- Financial Reports

Reports

- PDF Reports
 - Excel Reports
-

Out of Scope (Version 1.0)

The following features are excluded from the initial release and may be considered in future versions:

- AI-based equipment recommendations
 - Predictive maintenance using machine learning
 - IoT sensor integration
 - RFID equipment tracking
 - Voice assistant
 - Mobile applications (Android/iOS)
 - Offline synchronization
 - Blockchain-based audit logs
 - Multi-language support
 - Advanced workflow automation
-

8. Stakeholders

Stakeholder	Responsibility
Research Students	Equipment Booking
Faculty	Resource Utilization
Lab Technician	Equipment Maintenance
Lab Manager	Equipment Administration
Department Head	Department Oversight
Institution Administrator	Organization Management
Finance Department	Billing & Cost Allocation
System Administrator	Platform Administration

Development Team	Software Development
QA Team	Testing
DevOps Team	Deployment
Product Manager	Product Ownership

9. Assumptions

The following assumptions are considered during development:

- Every user belongs to one institution.
 - Each institution may contain multiple departments.
 - Each department may contain multiple laboratories.
 - Equipment belongs to one laboratory.
 - Every equipment item has a unique identifier.
 - Every booking requires equipment availability.
 - Only authorized users can approve bookings.
 - Notifications are delivered through integrated third-party services.
 - PostgreSQL is the primary transactional database.
 - Internet connectivity is available during platform usage.
-

10. Constraints

Technical Constraints

- Spring Boot backend.
- React frontend.
- PostgreSQL database.
- JWT authentication.
- Docker deployment.
- Cloud-ready architecture.

Business Constraints

- Project duration limited to **8 weeks**.
 - Internship scope should remain achievable.
 - System should support role-based security.
 - Reports must be exportable as PDF and Excel.
-

11. Success Criteria

The project will be considered successful if:

- All core modules are implemented.
 - Authentication is secure.
 - Equipment booking works end-to-end.
 - Maintenance workflows are functional.
 - Reports are generated successfully.
 - Dashboard displays accurate analytics.
 - Application is deployed successfully.
 - Role-based authorization functions correctly.
 - Average API response time remains under 2 seconds.
 - Documentation is complete.
-

12. Glossary

Term	Meaning
JWT	JSON Web Token used for secure authentication
OAuth2	Authentication protocol for third-party login
Equipment	Laboratory asset available for booking
Booking	Reservation of equipment for a specific time
Utilization	Percentage of equipment usage over time
Calibration	Process of verifying equipment accuracy
Maintenance	Repair or servicing of equipment
ROI	Return on Investment
Dashboard	Summary interface displaying KPIs and reports

 End of Part 1

This is the first section of a professional PRD.

Part 2 will cover:

- User Personas
- User Roles & Permission Matrix
- User Stories (50+)
- Acceptance Criteria
- Business Rules
- Detailed User Journey

From Part 2 onward, the document becomes much more detailed and implementation-oriented.

Excellent. Now Excellent. Now we'll continue with **Part 2** of the PRD.

This section is much closer to what product companies (Infosys, Microsoft, Atlassian, Zoho, etc.) write before development starts.

13. Target Users

The Lab Resource Utilization Platform is designed for multiple categories of users within universities, research institutions, laboratories, and collaborating organizations. Each user type has distinct responsibilities and system permissions.

13.1 Primary Users

1. Research Student

Description

Research students use laboratory equipment for academic research, experimentation, and project work.

Responsibilities

- Search laboratory equipment.
- View equipment availability.
- Request equipment bookings.
- View booking history.
- Receive booking notifications.
- Cancel bookings when required.

Goals

- Quickly locate available equipment.
- Book equipment without manual approvals where applicable.

- Receive reminders before scheduled usage.
 - Track previous equipment usage.
-

2. Faculty Member

Description

Faculty members supervise students and frequently reserve laboratory equipment for teaching and research activities.

Responsibilities

- Reserve equipment
- Approve student requests (department policy dependent)
- View departmental equipment usage
- Monitor research resource allocation

Goals

- Efficient scheduling
 - Better research planning
 - Reduced booking conflicts
-

3. Lab Technician

Description

Responsible for equipment maintenance and operational readiness.

Responsibilities

- Update equipment status
- Perform maintenance
- Record calibration details
- Upload maintenance reports
- Close maintenance requests

Goals

- Ensure equipment availability
 - Reduce downtime
 - Maintain maintenance records
-

4. Lab Manager

Description

Manages the complete laboratory and all equipment under a department.

Responsibilities

- Add new equipment
- Approve bookings
- Assign technicians
- View utilization reports
- Manage maintenance schedules

Goals

- Improve equipment utilization
 - Reduce idle equipment
 - Improve scheduling efficiency
-

5. Department Head

Description

Responsible for monitoring laboratory performance across departments.

Responsibilities

- Review reports
 - Monitor equipment utilization
 - Approve procurement recommendations
 - View department analytics
-

6. Institution Administrator

Description

Responsible for managing all laboratories within an institution.

Responsibilities

- Manage users
 - Manage departments
 - Configure institution
 - Generate organization reports
 - Monitor equipment inventory
-

7. System Administrator

Description

Platform owner responsible for overall application administration.

Responsibilities

- Manage institutions
 - Configure platform
 - Monitor system health
 - Audit logs
 - Security management
-

14. Secondary Users

Secondary users interact indirectly with the platform.

Finance Department

Responsibilities

- Generate invoices
 - Track usage costs
 - Budget allocation
 - Department billing
-

Maintenance Team

Responsibilities

- Work orders
 - Spare part management
 - Maintenance planning
-

External Research Institutions

Responsibilities

- Request shared equipment

- Manage external bookings
 - Track resource sharing
-

15. User Persona

Persona 1

Name

Arun Kumar

Role

Research Student

Age

23

Department

Mechanical Engineering

Goals

- Book CNC Machine
- Complete Final Year Project
- Receive booking reminders

Pain Points

- Doesn't know machine availability
- Manual booking takes time
- No booking history

Solution

Online booking platform

Persona 2

Name

Dr. Priya

Role

Faculty Member

Goals

- Reserve lab for practical sessions
- Approve student bookings

Pain Points

- Booking conflicts
- Manual approval

Solution

Department dashboard

Persona 3

Name

Rajesh

Role

Lab Technician

Goals

- Update maintenance
- Schedule calibration

Pain Points

- Paper maintenance logs

Solution

Maintenance dashboard

Persona 4

Name

Karthik

Role

Institution Admin

Goals

- Improve utilization
- Reduce procurement cost

Pain Points

- No centralized reports

Solution

Analytics Dashboard

16. Role & Permission Matrix

Feature	Student	Faculty	Technician	Manager	Dept Head	Institution Admin	System Admin
Login	✓	✓	✓	✓	✓	✓	✓
Register	✓	✓	✗	✗	✗	✗	✗
Search Equipment	✓	✓	✓	✓	✓	✓	✓
View Equipment	✓	✓	✓	✓	✓	✓	✓
Book Equipment	✓	✓	✗	✓	✗	✗	✗
Cancel Booking	✓	✓	✗	✓	✗	✗	✗

Approve Booking	✗	Optional	✗	✓	✓	✓	✓
Add Equipment	✗	✗	✗	✓	✗	✓	✓
Edit Equipment	✗	✗	✓	✓	✗	✓	✓
Delete Equipment	✗	✗	✗	✓	✗	✓	✓
Manage Users	✗	✗	✗	✗	✗	✓	✓
Manage Departments	✗	✗	✗	✗	✗	✓	✓
Manage Institutions	✗	✗	✗	✗	✗	✗	✓
View Reports	Limited	Department	Maintenance	Department	Department	Institution	Global
Generate Reports	✗	✗	✗	✓	✓	✓	✓
Configure System	✗	✗	✗	✗	✗	Limited	✓

17. User Stories

Authentication

US-001

As a user,

I want to log in securely

So that I can access my dashboard.

Acceptance Criteria

- Valid email required.
 - Valid password required.
 - JWT token generated.
 - Invalid credentials show error.
 - Redirect to dashboard after login.
-

Equipment Search

US-002

As a researcher,

I want to search laboratory equipment

So that I can locate available resources.

Acceptance Criteria

- Search by equipment name.
 - Search by category.
 - Search by department.
 - Search by institution.
 - Show equipment availability.
-

Booking Equipment

US-003

As a researcher,

I want to reserve equipment

So that I can perform my experiment.

Acceptance Criteria

- Equipment must be available.
 - Booking time should not overlap.
 - Confirmation notification sent.
 - Booking stored in database.
-

Equipment Approval

US-004

As a Lab Manager,

I want to approve bookings

So that equipment usage is controlled.

Acceptance Criteria

- View pending bookings.
 - Approve or reject.
 - Notify researcher.
 - Update booking status.
-

Maintenance

US-005

As a Technician,

I want to record maintenance

So that equipment history is maintained.

Acceptance Criteria

- Equipment status changes.
 - Maintenance log created.
 - Completion date stored.
 - Equipment becomes available after maintenance.
-

Analytics

US-006

As an Institution Admin,

I want utilization reports

So that procurement decisions can be made.

Acceptance Criteria

- Daily reports.
 - Monthly reports.
 - Equipment utilization.
 - Department comparison.
 - Export PDF.
-

18. Business Rules

The following business rules govern the application.

BR-001

A user can belong to only one institution.

BR-002

Every department belongs to one institution.

BR-003

Every laboratory belongs to one department.

BR-004

Equipment belongs to one laboratory.

BR-005

Equipment cannot be booked while under maintenance.

BR-006

Booking dates cannot overlap.

BR-007

Only managers can approve bookings.

BR-008

Only technicians can close maintenance tasks.

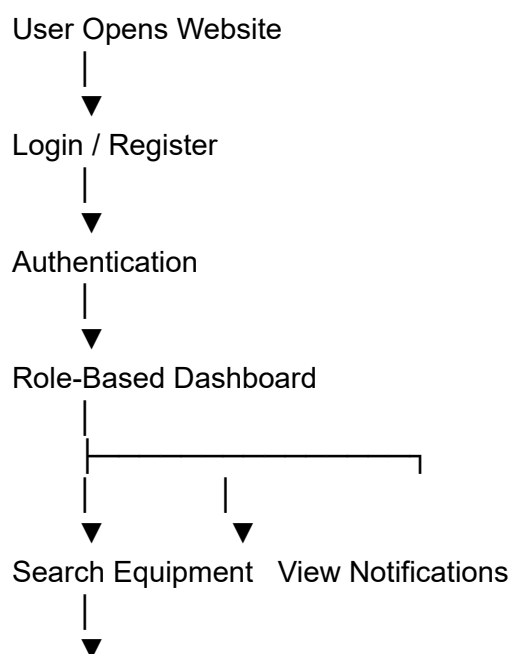
BR-009

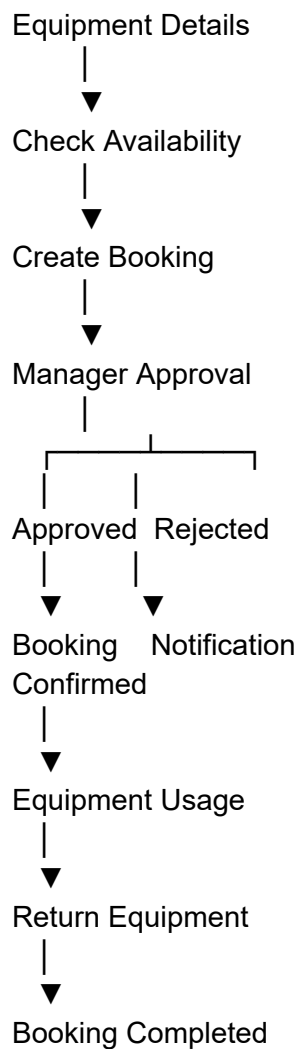
Deleted equipment cannot be booked.

BR-010

Every booking generates an audit log.

19. Functional Flow (High-Level User Journey)





End of Part 2

At this stage, the PRD has reached the level where a **UI/UX designer and backend architect** can begin planning. The next part will dive into **Module 1 (Authentication & Authorization)** with complete workflows, validations, database tables, APIs, screen details, sequence diagrams, and edge cases—essentially documenting it the way an enterprise team would before implementation.

20. Module 1 – Authentication & Authorization

20.1 Module Overview

Authentication is the first and most critical module of the Lab Resource Utilization Platform. It ensures that only authorized users can access the system while enforcing role-based permissions.

This module provides secure login, registration, JWT-based session management, OAuth2 login, password recovery, and user profile management.

The module is built using **Spring Security**, **JWT**, **OAuth2**, **BCrypt Password Encoder**, and **React Authentication Context**.

20.2 Objectives

The Authentication module should:

- Secure user credentials.
 - Authenticate users using JWT.
 - Authorize users based on assigned roles.
 - Protect all APIs.
 - Support Google OAuth Login.
 - Support Password Reset.
 - Maintain user sessions securely.
 - Prevent unauthorized access.
 - Record authentication logs.
-

20.3 Functional Scope

Included Features

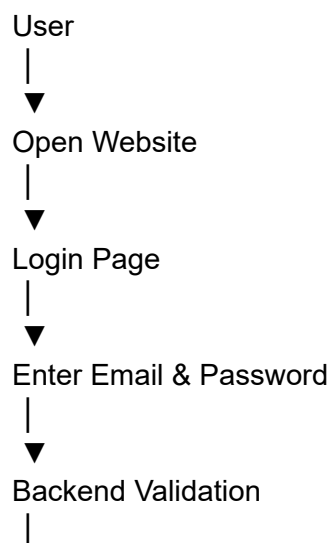
- ✓ User Registration
- ✓ Login
- ✓ JWT Authentication
- ✓ Google OAuth2 Login
- ✓ Forgot Password
- ✓ Reset Password
- ✓ Refresh Access Token

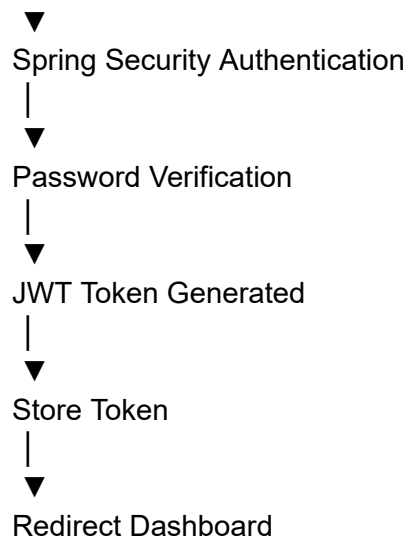
- ✓ Logout
 - ✓ Profile Management
 - ✓ Change Password
 - ✓ Role-Based Authorization
 - ✓ Session Validation
 - ✓ Authentication Logs
-

20.4 Actors

Actor	Access
Student	Register, Login
Faculty	Login
Lab Technician	Login
Lab Manager	Login
Institution Admin	Login
System Admin	Login

20.5 Authentication Workflow





20.6 Login Screen

Purpose

Allows registered users to access the application securely.

Components

Email

Password

Remember Me Checkbox

Forgot Password Link

Login Button

Google Login Button

Register Link

Validation Rules

Email required

Password required

Valid email format

Minimum password length

Maximum password length

Prevent SQL Injection

Prevent XSS

Rate Limiting

Success Flow

Login



Authentication Success



JWT Generated



Save Token



Dashboard

Failure Flow

Login



Invalid Credentials



Show Error



Retry Login

20.7 Registration Screen

Purpose

Allows new researchers to create accounts.

Fields

First Name

Last Name

Email

Phone Number

Institution

Department

Password

Confirm Password

Terms & Conditions

Validations

Email must be unique

Phone unique

Password minimum 8 characters

One uppercase

One lowercase

One number

One special character

Passwords must match

Institution mandatory

Department mandatory

Registration Flow

Fill Form



Validate



Save User



Encrypt Password



Send Verification Email



Account Created

20.8 Forgot Password Flow

Forgot Password



Enter Email



Verify Email Exists



Generate Reset Token

↓

Send Email

↓

User Opens Link

↓

Reset Password

↓

Login

20.9 Change Password Flow

Dashboard

↓

Profile

↓

Change Password

↓

Old Password

↓

New Password

↓

Confirm Password

↓

Update

↓

Logout

↓

Login Again

20.10 JWT Authentication Flow

Client

↓

Login API

↓

Spring Security

↓

Authentication Manager

↓

User Details Service

↓

Validate Password

↓

Generate JWT

↓

Return Token

↓

Client Stores Token

↓

API Requests



Authorization Header



JWT Validation



Access Granted

20.11 JWT Token Structure

Header

Payload

Signature

Payload Example

```
{  
  "userId":101,  
  "email":"student@abc.edu",  
  "role":"RESEARCHER",  
  "institutionId":5,  
  "exp":"2026-07-20T18:30:00"  
}
```

20.12 Role-Based Authorization

Role	Dashboard
Student	Research Dashboard
Faculty	Faculty Dashboard
Technician	Maintenance Dashboard

Manager	Lab Dashboard
Institution Admin	Institution Dashboard
System Admin	Global Dashboard

20.13 Screen Flow

Landing Page



Login



JWT Authentication



Role Verification



Dashboard

20.14 UI Components

Login Card

Email Textbox

Password Textbox

Remember Me

Google Login Button

Forgot Password Link

Login Button

Error Message

20.15 Database Tables

users

Field	Type
id	BIGINT
first_name	VARCHAR
last_name	VARCHAR
email	VARCHAR
phone	VARCHAR
password	VARCHAR
institution_id	BIGINT
department_id	BIGINT
role_id	BIGINT
status	BOOLEAN
created_at	TIMESTAMP

roles

Field	Type
id	BIGINT
role_name	VARCHAR

refresh_tokens

Field	Type
-------	------

id	BIGINT
user_id	BIGINT
token	TEXT
expiry_date	TIMESTAMP

password_reset_tokens

Field	Type
id	BIGINT
user_id	BIGINT
token	VARCHAR
expiry	TIMESTAMP

20.16 REST APIs

Register

POST /api/auth/register

Request

```
{
  "firstName": "Selva",
  "lastName": "K",
  "email": "selva@gmail.com",
  "password": "Password@123"
}
```

Response

```
{
  "message": "Registration Successful"
}
```

Login

POST /api/auth/login

Request

```
{  
  "email": "selva@gmail.com",  
  "password": "Password@123"  
}
```

Response

```
{  
  "accessToken": "JWT_TOKEN",  
  "refreshToken": "REFRESH_TOKEN",  
  "role": "LAB_MANAGER"  
}
```

Forgot Password

POST /api/auth/forgot-password

Reset Password

POST /api/auth/reset-password

Refresh Token

POST /api/auth/refresh

Logout

POST /api/auth/logout

20.17 Exception Handling

Error	HTTP Code
Invalid Credentials	401
User Not Found	404
Email Exists	409
Invalid Token	401

Expired Token	401
Validation Failed	400
Internal Error	500

20.18 Business Rules

- Email must be unique.
 - Phone number must be unique.
 - Password stored only in encrypted form (BCrypt).
 - JWT expires after configured duration (e.g., 1 hour).
 - Refresh token required after access token expiry.
 - Locked users cannot log in.
 - Inactive users cannot log in.
 - Only System Admin can assign Institution Admin roles.
 - Every login and logout action should be recorded in the audit log.
-

20.19 Acceptance Criteria

ID	Criteria
AC-001	User can register with valid details
AC-002	Duplicate email is rejected
AC-003	Password is encrypted before storage
AC-004	Valid users receive JWT and Refresh Token
AC-005	Invalid credentials return HTTP 401
AC-006	Protected APIs require a valid JWT
AC-007	Role-based access is enforced
AC-008	Password reset works using secure tokens
AC-009	User can update profile details
AC-010	Audit log records authentication events

20.20 Future Enhancements

- Multi-Factor Authentication (MFA)
 - Biometric Authentication
 - Single Sign-On (SSO)
 - Microsoft Azure AD Login
 - LDAP Integration
 - Device Management
 - Login History Dashboard
 - Session Management Console
-

End of Part 3

This module is now documented in enough detail that both frontend and backend developers can begin implementation.

Next Part (Part 4)

We'll document **Module 2 – Institution Management**, including:

- Institution hierarchy (Institution → Department → Lab)
- Database schema
- CRUD operations
- Screen flows
- APIs
- Validation rules
- User interface details
- Business rules
- Acceptance criteria
- Sequence diagrams
- Edge cases

This level of detail is what turns the PRD into a practical development blueprint rather than just a high-level requirements document.

Excellent. Now we're entering the **core business module** of the project.

This module is the **heart of the application** because every other module (Booking, Maintenance, Analytics, Billing, Reports) depends on it.

A real Product Manager would spend **15–20 pages** documenting this module.

21. Module 2 – Institution Management

21.1 Module Overview

The Institution Management module forms the organizational backbone of the Lab Resource Utilization Platform. It enables the creation and management of institutions, departments, laboratories, and their relationships.

This module provides a hierarchical organizational structure where:

- One System manages multiple Institutions.
- One Institution contains multiple Departments.
- One Department contains multiple Laboratories.
- One Laboratory contains multiple Equipment assets.

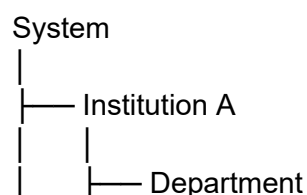
This hierarchy ensures that all resources, users, bookings, and reports are properly organized and managed.

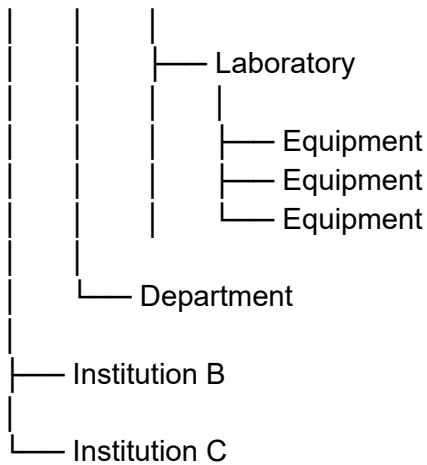
21.2 Module Objectives

The Institution Management module should:

- Allow the System Administrator to manage institutions.
 - Allow Institution Administrators to manage departments.
 - Allow Department Heads to manage laboratories.
 - Associate equipment with laboratories.
 - Associate users with departments.
 - Support institution-specific policies.
 - Provide organization-wide reporting.
 - Enable inter-institution resource sharing.
-

21.3 Organization Hierarchy





21.4 Features

Institution Management

Features

- Create Institution
 - Edit Institution
 - Delete Institution
 - Activate Institution
 - Deactivate Institution
 - View Institution Details
 - Institution Search
 - Institution Logo Upload
 - Institution Address Management
-

Department Management

Features

- Create Department
 - Edit Department
 - Delete Department
 - Assign Department Head
 - Department Status
 - Department Search
-

Laboratory Management

Features

- Create Laboratory
- Edit Laboratory
- Delete Laboratory
- Assign Lab Manager
- Assign Equipment
- Laboratory Status

User Assignment

Features

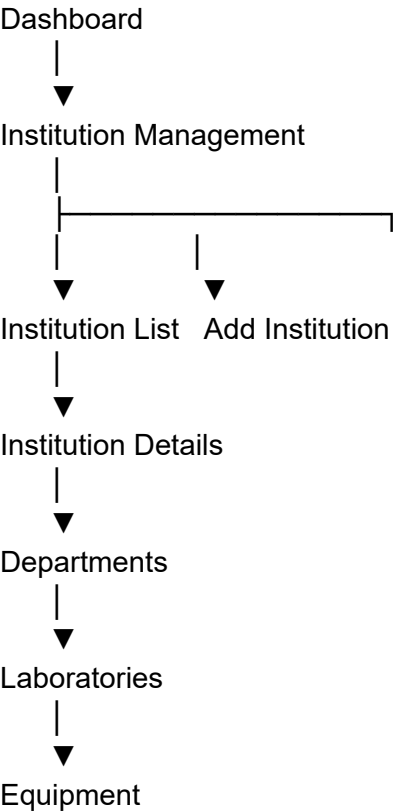
- Assign User to Institution
- Assign User to Department
- Assign User to Laboratory
- Transfer User
- Remove User

21.5 User Access Matrix

Operation	System Admin	Institution Admin	Department Head	Lab Manager
Create Institution	✓	✗	✗	✗
Update Institution	✓	✗	✗	✗
Delete Institution	✓	✗	✗	✗
Create Department	✓	✓	✗	✗
Edit Department	✓	✓	✗	✗
Delete Department	✓	✓	✗	✗
Create Laboratory	✓	✓	✓	✗

Update Laboratory	✓	✓	✓	✗
Delete Laboratory	✓	✓	✓	✗
Assign Users	✓	✓	✓	✗

21.6 Screen Flow



21.7 Screen Inventory

Institution Module

Screen IM-01

Institution List

Purpose

Displays all registered institutions.

Components

- Search Box
- Institution Table
- Status Filter
- Add Institution Button
- Edit Button
- Delete Button
- Pagination

Actions

- View
 - Edit
 - Delete
 - Activate
 - Deactivate
-

Screen IM-02

Add Institution

Fields

- Institution Name
- Institution Code
- Email
- Contact Number
- Website
- Address
- City
- State
- Country
- PIN Code
- Logo Upload

Buttons

- Save
 - Reset
 - Cancel
-

Screen IM-03

Institution Details

Displays

- Institution Information
 - Departments
 - Laboratories
 - Equipment Count
 - User Count
 - Reports
-

Screen IM-04

Edit Institution

Allows updating all institution information.

Screen IM-05

Department Management

Displays

Department List

Actions

- Add
 - Edit
 - Delete
-

Screen IM-06

Laboratory Management

Displays

Laboratory List

Actions

- Add
- Edit
- Delete

21.8 UI Components

Institution Card

Department Card

Lab Card

Breadcrumb Navigation

Search Bar

Filter Dropdown

Data Table

Modal Dialog

Confirmation Popup

Toast Notification

21.9 Validation Rules

Institution

Institution Name

- Required
- Minimum 3 characters
- Maximum 150 characters
- Must be unique

Institution Code

- Required
- Unique

Email

- Valid email format
- Unique

Phone

- Valid number
- Unique

Website

- Valid URL
-

Department

Department Name

Required

Unique within Institution

Laboratory

Laboratory Name

Required

Unique within Department

21.10 Database Tables

institutions

Field	Type
id	BIGINT
institution_code	VARCHAR
institution_name	VARCHAR
email	VARCHAR
phone	VARCHAR
website	VARCHAR

address	TEXT
city	VARCHAR
state	VARCHAR
country	VARCHAR
pincode	VARCHAR
logo_url	VARCHAR
status	BOOLEAN
created_at	TIMESTAMP

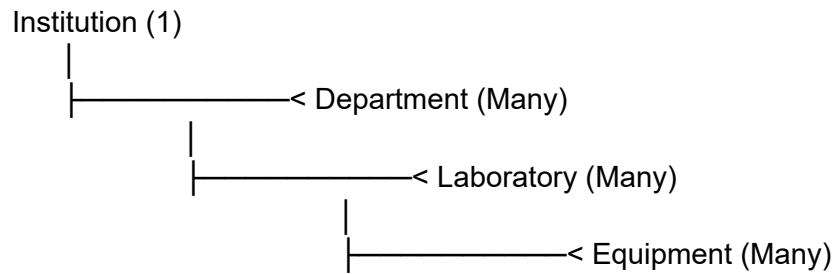
departments

Field	Type
id	BIGINT
institution_id	BIGINT
department_name	VARCHAR
hod_user_id	BIGINT
status	BOOLEAN

laboratories

Field	Type
id	BIGINT
department_id	BIGINT
laboratory_name	VARCHAR
lab_manager_id	BIGINT
location	VARCHAR
status	BOOLEAN

21.11 Entity Relationships



21.12 REST APIs

Institution APIs

Create Institution

POST /api/institutions

Get All Institutions

GET /api/institutions

Get Institution By ID

GET /api/institutions/{id}

Update Institution

PUT /api/institutions/{id}

Delete Institution

DELETE /api/institutions/{id}

Department APIs

POST /api/departments

GET /api/departments

PUT /api/departments/{id}

DELETE /api/departments/{id}

Laboratory APIs

POST /api/laboratories

GET /api/laboratories

PUT /api/laboratories/{id}

DELETE /api/laboratories/{id}

21.13 Business Rules

BR-011

Institution Code must be unique.

BR-012

Institution cannot be deleted if departments exist.

BR-013

Department cannot be deleted if laboratories exist.

BR-014

Laboratory cannot be deleted if equipment exists.

BR-015

Every laboratory must belong to one department.

BR-016

Every department must belong to one institution.

BR-017

Only active institutions allow bookings.

BR-018

Institution Admin cannot modify another institution.

BR-019

Only System Admin can delete institutions.

BR-020

Users cannot belong to multiple institutions simultaneously.

21.14 Sequence Diagram

System Admin

|

Create Institution

|

React UI

|

Institution Controller

|

Institution Service

|

Validation

|

Save Institution

|

PostgreSQL

|

Success Response

|

Dashboard Updated

21.15 Exception Handling

Scenario	HTTP Status
Duplicate Institution Code	409 Conflict
Institution Not Found	404 Not Found
Validation Failed	400 Bad Request
Unauthorized	401 Unauthorized
Forbidden	403 Forbidden
Internal Server Error	500 Internal Server Error

21.16 Acceptance Criteria

ID	Requirement
AC-021	System Admin can create institutions

- AC-022 Institution code must be unique
 - AC-023 Institution Admin can manage departments
 - AC-024 Department Head can manage laboratories
 - AC-025 Laboratory hierarchy is maintained
 - AC-026 Duplicate institutions are prevented
 - AC-027 Institution deletion follows business rules
 - AC-028 Organization hierarchy displays correctly
 - AC-029 APIs enforce role-based access
 - AC-030 Audit logs are generated for create, update, and delete operations
-

21.17 Future Enhancements

- Multi-campus institution support
 - Institution branding and themes
 - Geographic location mapping
 - GIS integration for campus maps
 - Bulk import/export of departments and laboratories
 - Organization hierarchy visualization
 - Institution-level quotas and licensing
-

End of Part 4

At this stage, the PRD now covers:

- ☒ Project Overview
 - ☒ Stakeholders and Business Goals
 - ☒ User Personas and Role Matrix
 - ☒ Authentication Module (implementation-level)
 - ☒ Institution Management Module (implementation-level)
-

Next (Part 5): Equipment Inventory Management

This is the **largest module** in the project (around 15–20 pages). It will cover:

- Equipment lifecycle
- Equipment categories and metadata
- QR code management
- Image and document uploads
- Availability tracking
- Inventory workflows
- Screen-by-screen UI
- Database schema
- APIs
- Business rules
- Validation rules
- Acceptance criteria
- Equipment state transitions

This module forms the core of the platform and will drive the booking, maintenance, and analytics features.

22. Module 3 – Equipment Inventory Management

22.1 Module Overview

The Equipment Inventory Management Module is responsible for managing all laboratory equipment across institutions. It provides a centralized repository where administrators can register, organize, categorize, search, update, and monitor equipment throughout its lifecycle.

Every equipment item acts as an organizational asset and is uniquely identified within the platform. The module supports tracking equipment specifications, ownership, operational status, maintenance history, calibration records, location, documentation, and availability.

The module is the foundation for:

- Equipment Booking
- Resource Sharing
- Maintenance
- Billing
- Analytics
- Reports

Without this module, none of the remaining modules can function.

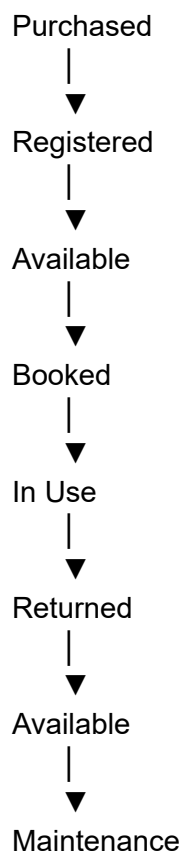
22.2 Module Objectives

The Equipment Inventory Module shall:

- Maintain a centralized inventory of all laboratory equipment.
 - Support equipment categorization.
 - Store equipment specifications and technical information.
 - Track equipment availability.
 - Track equipment ownership.
 - Maintain calibration records.
 - Store equipment manuals and documents.
 - Generate QR Codes.
 - Support advanced searching.
 - Maintain equipment lifecycle history.
-

22.3 Equipment Lifecycle

Every equipment passes through multiple stages.





22.4 Equipment Status

Status	Description
Available	Ready for Booking
Reserved	Booking Approved
In Use	Currently Used
Under Maintenance	Technician Working
Calibration Due	Calibration Required
Out of Service	Equipment Failure
Retired	Permanently Removed

22.5 Equipment Categories

The platform should support unlimited categories.

Examples

- Mechanical
- Electrical
- Electronics
- Computer Science
- Biomedical
- Civil Engineering
- Chemical Engineering
- Physics
- Chemistry
- Biology

Each category may contain multiple equipment.

22.6 Equipment Master Information

Each equipment should contain the following information.

Basic Details

Equipment Name

Equipment Code

Serial Number

Manufacturer

Model Number

Category

Sub Category

Brand

Description

Purchase Date

Purchase Cost

Warranty Period

Current Value

Asset Number

QR Code

Barcode

Location Details

Institution

Department

Laboratory

Building

Floor

Room Number

Rack Number

GPS Location (Future)

Operational Details

Current Status

Available From

Maximum Booking Hours

Power Requirement

Operating Manual

Calibration Due Date

Maintenance Schedule

Assigned Technician

Ownership Details

Purchased By

Funding Source

Grant Number

Vendor

Invoice Number

AMC Provider

AMC Expiry

22.7 User Roles

Role	Access
Researcher	View Equipment
Faculty	View Equipment
Technician	Update Status
Lab Manager	Full CRUD
Institution Admin	Manage Inventory
System Admin	Global Inventory

22.8 Equipment Registration Workflow

Dashboard



Equipment Management



Add Equipment



Fill Details



Upload Documents



Generate QR Code



Save Equipment



Equipment Registered



Available for Booking

22.9 Equipment Search Workflow

Equipment List



Search



Filter



Sort



Equipment Details

22.10 Equipment Details Screen

Purpose

Displays complete equipment information.

Sections

General Information

Technical Information

Availability

Booking History

Maintenance History

Calibration

Documents

Analytics

Components

Equipment Image

Equipment Name

Status Badge

QR Code

Availability Calendar

Recent Bookings

Recent Maintenance

Download Manual

Book Button

22.11 Equipment List Screen

Purpose

Displays all available equipment.

Features

Search

Pagination

Filters

Sorting

Status Badges

Category Filter

Department Filter

Institution Filter

Export

Bulk Selection

22.12 Add Equipment Screen

Fields

Equipment Name

Equipment Code

Category

Manufacturer

Model Number

Purchase Date

Purchase Cost

Department

Laboratory

Technician

Warranty

Manual Upload

Image Upload

QR Code

Buttons

Save

Reset

Cancel

22.13 Edit Equipment Screen

Allows editing

Specifications

Location

Status

Documents

Images

Warranty

Technician

22.14 Equipment Availability Calendar

Displays

Daily Availability

Weekly Availability

Monthly Availability

Booked Slots

Maintenance Slots

Blocked Slots

Available Slots

22.15 Equipment Documents

Supported Files

PDF

DOCX

Images

Calibration Certificate

Warranty Card

Invoice

Operating Manual

Safety Guidelines

22.16 QR Code Workflow

Equipment Created



Generate QR Code



Store QR Image



Print Sticker



Attach to Equipment



Scan



Equipment Details

22.17 Database Tables

equipment

Column	Type
id	BIGINT
equipment_code	VARCHAR
equipment_name	VARCHAR
category_id	BIGINT
laboratory_id	BIGINT
manufacturer	VARCHAR
model_number	VARCHAR
serial_number	VARCHAR
purchase_date	DATE
purchase_cost	DECIMAL
warranty_expiry	DATE
status	VARCHAR
qr_code	VARCHAR
image_url	VARCHAR
created_at	TIMESTAMP

equipment_category

Column	Type
id	BIGINT
category_name	VARCHAR
description	TEXT

equipment_documents

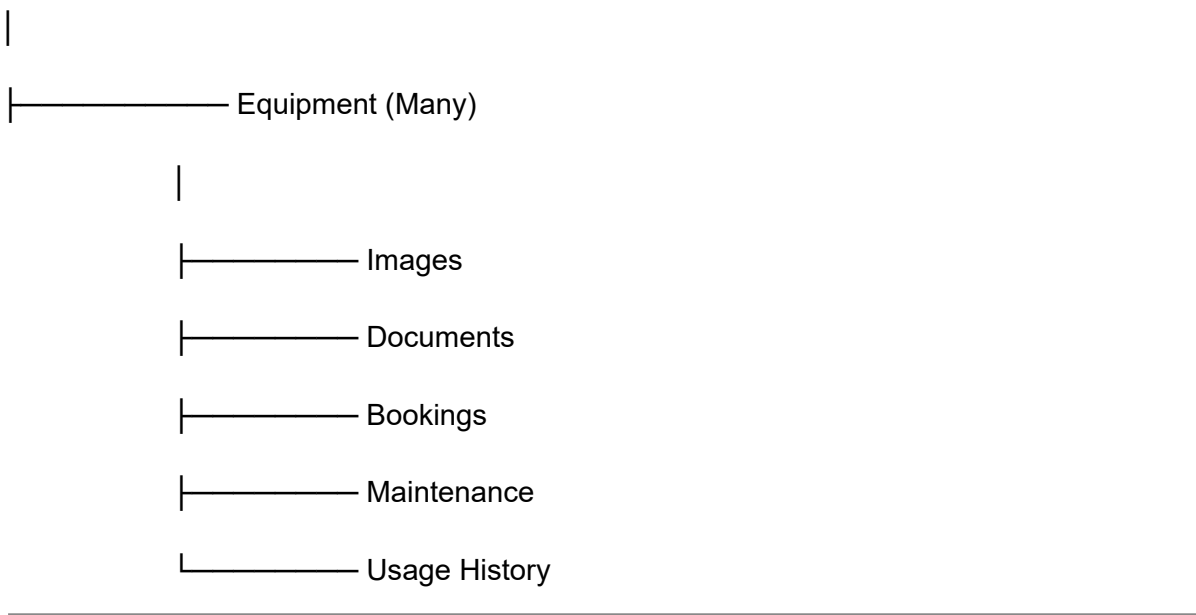
Column	Type
id	BIGINT
equipment_id	BIGINT
file_name	VARCHAR
document_type	VARCHAR
uploaded_at	TIMESTAMP

equipment_images

Column	Type
id	BIGINT
equipment_id	BIGINT
image_url	VARCHAR

22.18 Entity Relationship

Category (1)



22.19 REST APIs

Equipment APIs

Create Equipment

POST /api/equipment

Get Equipment

GET /api/equipment

Equipment Details

GET /api/equipment/{id}

Update Equipment

PUT /api/equipment/{id}

Delete Equipment

DELETE /api/equipment/{id}

Equipment Search

GET /api/equipment/search

Query Parameters

name

category

department

institution

status

Upload Equipment Image

POST /api/equipment/{id}/image

Upload Manual

POST /api/equipment/{id}/manual

Generate QR Code

POST /api/equipment/{id}/qr

22.20 Validation Rules

Equipment Name

Required

Maximum 200 characters

Equipment Code

Unique

Required

Purchase Cost

Must be positive

Purchase Date

Cannot be future date

Warranty Date

Must be greater than purchase date

Department

Required

Laboratory

Required

Image

Maximum 5MB

22.21 Business Rules

BR-021

Equipment Code must be unique.

BR-022

Equipment cannot exist without a laboratory.

BR-023

Deleted equipment cannot be restored.

BR-024

Only Available equipment can be booked.

BR-025

Equipment under maintenance cannot accept bookings.

BR-026

QR Code must be generated for every equipment.

BR-027

Equipment image is mandatory before publishing.

BR-028

Equipment documents must be version-controlled.

BR-029

Equipment category cannot be deleted if equipment exists.

BR-030

Retired equipment is read-only and cannot be modified.

22.22 Exception Handling

Exception	HTTP
Equipment Not Found	404
Duplicate Equipment Code	409
Invalid Category	400
Invalid Laboratory	400
Unauthorized	401
Forbidden	403
File Upload Failed	500

22.23 Acceptance Criteria

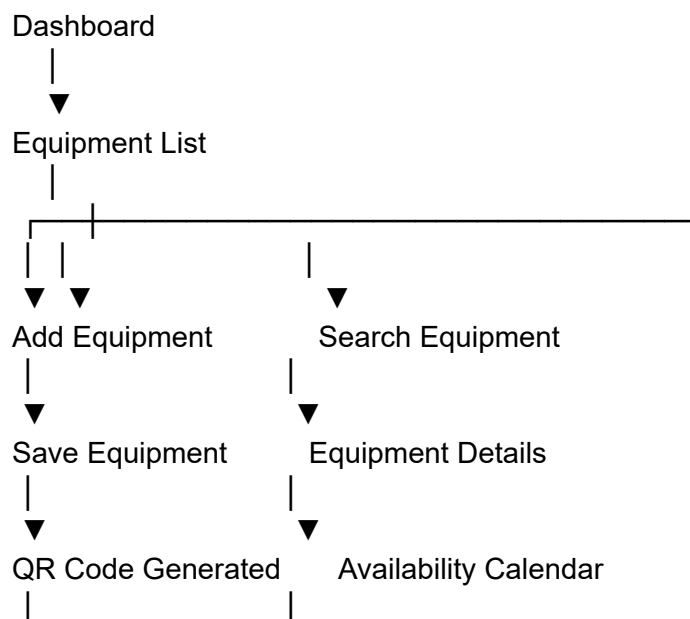
ID	Requirement
AC-031	Lab Manager can add equipment
AC-032	Equipment Code is unique
AC-033	QR Code is generated
AC-034	Equipment image uploads successfully

- AC-035 Documents upload successfully
 - AC-036 Equipment search works with filters
 - AC-037 Status changes are reflected immediately
 - AC-038 Equipment details display correctly
 - AC-039 CRUD operations are audited
 - AC-040 Only authorized roles can modify inventory
-

22.24 Future Enhancements

- RFID-based asset tracking
 - IoT sensor integration for live status
 - AI-powered equipment recommendations
 - Automatic depreciation calculation
 - Predictive replacement planning
 - Digital twin visualization
 - Barcode scanner integration
 - Equipment health score
 - Spare parts inventory management
 - Vendor portal integration
-

22.25 Screen Flow



22.26 Screen Inventory

Screen ID	Screen Name
EQ-01	Equipment List
EQ-02	Equipment Details
EQ-03	Add Equipment
EQ-04	Edit Equipment
EQ-05	Equipment Calendar
EQ-06	Equipment Documents
EQ-07	Equipment Images
EQ-08	QR Code Preview
EQ-09	Equipment Analytics
EQ-10	Equipment History

End of Part 5

At this point, your PRD has become a true enterprise document. We've fully specified the inventory module with workflows, data model, business rules, APIs, screens, and acceptance criteria.

Next (Part 6): Booking & Scheduling Module

This is the most complex functional module. We'll define:

- Booking workflow
- Calendar scheduling
- Approval/rejection process
- Waitlist management
- Booking conflict detection
- Recurring bookings
- Calendar UI
- Booking APIs

- Database schema
- State transitions
- Business rules
- Edge cases
- Notifications
- Sequence diagrams

This section will effectively describe how the platform manages equipment reservations from start to finish.

Excellent. We're now moving into the **intelligence layer** of the application. Unlike previous modules that focused on CRUD operations and workflows, this module provides **real-time insights and decision support** for administrators, lab managers, and institution leaders.

24. Module 5 – Real-Time Utilization Monitoring & Analytics

24.1 Module Overview

The Real-Time Utilization Monitoring & Analytics module collects, processes, and visualizes operational data from equipment bookings, maintenance activities, and user interactions. It transforms raw data into actionable insights, enabling institutions to optimize equipment usage, reduce idle time, plan future investments, and improve operational efficiency.

The module provides dashboards, KPIs, utilization trends, heatmaps, and predictive metrics for stakeholders at different organizational levels.

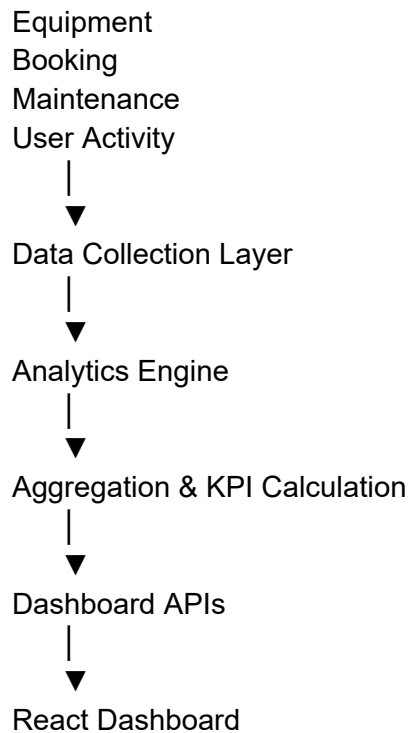
24.2 Module Objectives

The Analytics Module shall:

- Monitor real-time equipment utilization.
- Calculate utilization percentages.
- Detect idle equipment.
- Analyze booking trends.
- Compare departmental usage.
- Generate KPI dashboards.

- Provide downloadable reports.
 - Support decision-making for procurement.
 - Track maintenance impact on availability.
 - Offer predictive insights (future enhancement).
-

24.3 Analytics Architecture



24.4 Key Performance Indicators (KPIs)

KPI	Description
Equipment Utilization %	Percentage of booked time vs. available time
Idle Equipment	Equipment not used within a configurable period
Total Bookings	Number of completed and active bookings
Booking Success Rate	Approved bookings ÷ Total booking requests
Booking Rejection Rate	Rejected bookings ÷ Total booking requests
Maintenance Downtime	Total hours equipment was unavailable
Average Booking Duration	Mean reservation duration

Peak Usage Hours	Time periods with highest demand
Equipment Availability	Percentage of time equipment is available
Top Used Equipment	Most frequently reserved assets

24.5 User Roles

Role	Dashboard Access
Researcher	Personal Usage Dashboard
Faculty	Department Summary
Lab Manager	Laboratory Dashboard
Department Head	Department Analytics
Institution Admin	Institution Dashboard
System Admin	Global Dashboard

24.6 Dashboard Overview

The dashboard is divided into six major sections.

Dashboard Components

- KPI Cards
 - Utilization Charts
 - Equipment Status Summary
 - Booking Trends
 - Maintenance Statistics
 - Alerts & Notifications
-

24.7 KPI Cards

Each dashboard displays summary cards such as:

- Total Equipment
 - Active Equipment
 - Available Equipment
 - Equipment Under Maintenance
 - Active Bookings
 - Pending Approvals
 - Completed Bookings
 - Utilization Percentage
 - Idle Equipment Count
 - Upcoming Maintenance
-

24.8 Utilization Dashboard

Displays:

- Daily Utilization
- Weekly Utilization
- Monthly Utilization
- Quarterly Utilization
- Yearly Utilization

Metrics:

- Booking Hours
- Available Hours
- Utilization %

Formula:

Utilization % =
 $(\text{Total Booked Hours} \div \text{Total Available Hours}) \times 100$

Example:

Available Hours = 200

Booked Hours = 150

Utilization =

$150 / 200 \times 100$

= 75%

24.9 Equipment Heatmap

Purpose

Visualize periods of high and low equipment usage.

Legend

- Green → Low Usage
- Yellow → Medium Usage
- Orange → High Usage
- Red → Very High Usage

Example:

Monday

09:00 

10:00 

11:00 

12:00 

24.10 Booking Trend Analysis

Reports include:

- Daily bookings
- Weekly bookings
- Monthly bookings
- Semester trends
- Year-over-year comparison

Metrics

- New bookings
 - Completed bookings
 - Cancelled bookings
 - Rejected bookings
-

24.11 Equipment Performance Dashboard

Displays:

- Top 10 Most Used Equipment
 - Least Used Equipment
 - Frequently Cancelled Equipment
 - Equipment with Highest Downtime
 - Equipment Awaiting Maintenance
-

24.12 Department Analytics

Each department dashboard shows:

- Equipment Count
- Booking Count
- Utilization %
- Maintenance Hours
- Average Booking Duration
- Equipment Availability

Department comparison example:

Department	Utilization
Mechanical	88%
Electronics	74%
Computer Science	69%
Civil	53%

24.13 Institution Analytics

Displays:

- Institution-wise equipment distribution
- Institution utilization

- Active users
 - Total laboratories
 - Resource sharing statistics
 - Department rankings
-

24.14 Alerts & Notifications

Automatic alerts generated for:

- Equipment idle for more than 30 days
 - Utilization below threshold
 - Calibration due soon
 - Maintenance overdue
 - Booking rejection spike
 - Low equipment availability
 - Resource over-utilization
-

24.15 Screen Inventory

Screen ID	Screen Name
AN-01	Analytics Dashboard
AN-02	Utilization Dashboard
AN-03	Equipment Performance
AN-04	Department Analytics
AN-05	Institution Analytics
AN-06	Booking Trends
AN-07	KPI Dashboard
AN-08	Reports & Export

24.16 Database Views / Aggregated Tables

equipment_utilization_summary

Column	Type
equipment_id	BIGINT
booked_hours	DECIMAL
available_hours	DECIMAL
utilization_percentage	DECIMAL
last_updated	TIMESTAMP

booking_statistics

Column	Type
booking_date	DATE
total_bookings	INT
approved	INT
rejected	INT
completed	INT
cancelled	INT

department_utilization

Column	Type
department_id	BIGINT
utilization_percentage	DECIMAL
total_equipment	INT
booking_hours	DECIMAL

24.17 REST APIs

Dashboard Summary

GET /api/analytics/dashboard

Equipment Utilization

GET /api/analytics/utilization

Department Analytics

GET /api/analytics/departments

Institution Analytics

GET /api/analytics/institutions

Booking Trends

GET /api/analytics/bookings

Export Analytics Report

GET /api/analytics/export

Query Parameters:

format=pdf | excel | csv

dateRange=start,end

departmentId

institutionId

24.18 Validation Rules

- Date range must be valid.
 - End date cannot be earlier than start date.
 - Department must belong to selected institution.
 - Export format must be supported.
 - Users can only access analytics permitted by their role.
-

24.19 Business Rules

BR-041

Utilization is calculated only from completed and active bookings.

BR-042

Maintenance hours are excluded from available hours.

BR-043

Idle equipment is defined as no completed bookings for the configured threshold (default: 30 days).

BR-044

Department Heads can view only their department analytics.

BR-045

Institution Administrators cannot access analytics for other institutions.

BR-046

Reports are generated using the latest available aggregated data.

BR-047

Dashboard data refreshes automatically at configurable intervals (e.g., every 5 minutes).

BR-048

Analytics calculations must ignore deleted or retired equipment unless historical reporting is requested.

BR-049

Exports include only data the requesting user is authorized to access.

BR-050

Every report generation is logged for audit purposes.

24.20 Exception Handling

Exception	HTTP Status
Invalid Date Range	400
Unauthorized Access	401
Forbidden Analytics Scope	403
Department Not Found	404

Institution Not Found 404

Export Generation Failed 500

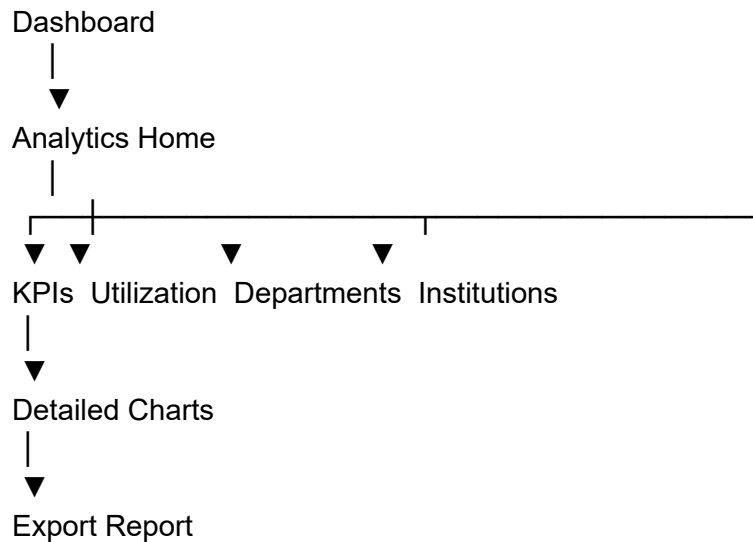
24.21 Acceptance Criteria

ID	Requirement
AC-051	Dashboard displays KPI cards correctly
AC-052	Utilization percentage is calculated accurately
AC-053	Heatmap reflects booking activity
AC-054	Department analytics display correct metrics
AC-055	Institution analytics aggregate departmental data
AC-056	Reports export successfully in supported formats
AC-057	Idle equipment alerts trigger correctly
AC-058	Dashboard respects role-based access
AC-059	Auto-refresh updates dashboard data
AC-060	Audit logs record report generation

24.22 Future Enhancements

- AI-powered demand forecasting
 - Predictive maintenance recommendations
 - Equipment utilization forecasting
 - Cost optimization suggestions
 - Carbon footprint analysis
 - Digital twin dashboards
 - Machine learning anomaly detection
 - Interactive drill-down dashboards
 - Real-time IoT sensor integration
 - Executive mobile analytics dashboard
-

24.23 Screen Flow



✓ End of Part 7

The PRD now includes a complete analytics module with KPI definitions, dashboards, reporting, data aggregation, APIs, business rules, and acceptance criteria.

Next (Part 8): Maintenance & Calibration Management

This module will document:

- Preventive and corrective maintenance workflows
- Calibration scheduling
- Work order management
- Technician assignment
- Spare parts tracking
- Maintenance history
- Downtime calculations
- APIs, database schema, UI screens, validations, business rules, and acceptance criteria

This module ensures equipment reliability and regulatory compliance throughout its operational lifecycle.

Excellent. This module distinguishes the platform from a standard Laboratory Management System (LMS). It enables **multiple universities, research institutions, and laboratories** to securely share high-value equipment, maximizing utilization while reducing procurement costs.

26. Module 7 – Inter-Institution Resource Sharing

26.1 Module Overview

The Inter-Institution Resource Sharing module enables registered institutions to share laboratory equipment with partner institutions through a secure and policy-driven workflow. It supports cross-institution equipment discovery, booking requests, approval workflows, cost allocation, usage tracking, and access control.

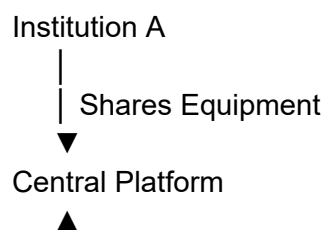
The module promotes collaboration by allowing institutions to use specialized equipment that they may not own, thereby improving utilization and reducing duplicate investments.

26.2 Module Objectives

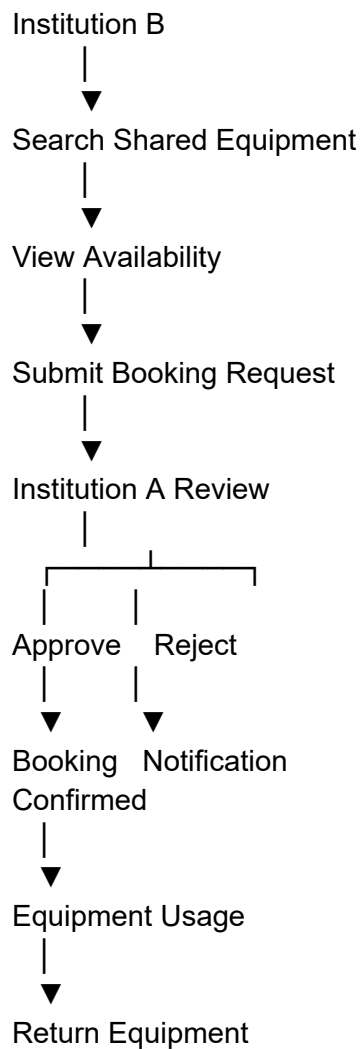
The Resource Sharing Module shall:

- Enable cross-institution equipment sharing.
 - Support institution partnership management.
 - Allow external booking requests.
 - Enforce sharing policies.
 - Manage approval workflows.
 - Allocate usage costs.
 - Track shared equipment usage.
 - Synchronize availability across institutions.
 - Generate collaboration reports.
 - Maintain complete audit trails.
-

26.3 Resource Sharing Model



26.4 Sharing Workflow



26.5 Sharing Policies

Each institution can configure:

- Public Sharing
- Partner-Only Sharing
- Department-Restricted Sharing
- Approval Required
- Hourly Usage Charges

- Daily Usage Charges
 - Security Deposit
 - Maximum Booking Duration
 - Booking Notice Period
 - Cancellation Policy
-

26.6 User Roles

Role	Responsibilities
External Researcher	Search and request shared equipment
Institution Admin	Configure sharing policies
Lab Manager	Approve or reject external requests
Finance Officer	Review usage charges
System Admin	Manage institution partnerships

26.7 Institution Partnership Management

Administrators can:

- Create partnership agreements.
 - Invite partner institutions.
 - Accept or reject partnership requests.
 - Define agreement validity.
 - Configure equipment categories available for sharing.
 - Suspend or terminate partnerships.
-

26.8 Shared Equipment Catalog

Each shared equipment entry includes:

- Equipment Name
- Equipment Code
- Category
- Institution

- Laboratory
 - Availability Status
 - Booking Calendar
 - Usage Cost
 - Security Deposit
 - Operating Restrictions
 - Required Training
 - Contact Person
-

26.9 Screens

RS-01 – Shared Equipment Search

Features:

- Search
 - Advanced Filters
 - Institution Filter
 - Category Filter
 - Availability Filter
 - Cost Filter
-

RS-02 – Equipment Details

Displays:

- Specifications
 - Availability Calendar
 - Institution Information
 - Usage Charges
 - Booking Button
 - Documents
 - Safety Instructions
-

RS-03 – External Booking Request

Fields:

- Equipment

- Institution
- Research Purpose
- Booking Date
- Start Time
- End Time
- Principal Investigator
- Funding Source
- Remarks

Buttons:

- Submit
 - Cancel
-

RS-04 – Approval Dashboard

Displays:

- Pending Requests
 - Institution Name
 - Requested Duration
 - Cost Estimate
 - Approve
 - Reject
 - Add Comments
-

RS-05 – Partnership Management

Displays:

- Active Partners
 - Pending Invitations
 - Agreement Expiry
 - Shared Categories
 - Status
-

26.10 Cost Allocation

The system supports:

- Free sharing
- Hourly billing
- Daily billing
- Fixed booking charges
- Security deposits
- Damage recovery costs

Example:

Usage Time = 5 Hours

Hourly Rate = ₹500

Total Charge

= 5 × 500

= ₹2500

26.11 Availability Synchronization

Whenever equipment is:

- Booked
- Under maintenance
- Reserved
- Retired

its availability is synchronized across all partner institutions to prevent conflicting reservations.

26.12 Database Tables

institution_partnerships

Column	Type
id	BIGINT
institution_a_id	BIGINT
institution_b_id	BIGINT

agreement_start	DATE
agreement_end	DATE
status	VARCHAR

shared_equipment

Column	Type
id	BIGINT
equipment_id	BIGINT
hourly_rate	DECIMAL
daily_rate	DECIMAL
security_deposit	DECIMAL
sharing_status	VARCHAR

external_booking_requests

Column	Type
id	BIGINT
shared_equipment_id	BIGINT
requesting_institution_id	BIGINT
requested_by	BIGINT
booking_date	DATE
start_time	TIME
end_time	TIME
status	VARCHAR

sharing_transactions

Column	Type
--------	------

id	BIGINT
booking_request_id	BIGINT
total_amount	DECIMAL
payment_status	VARCHAR
created_at	TIMESTAMP

26.13 REST APIs

Search Shared Equipment

GET /api/shared-equipment

Equipment Details

GET /api/shared-equipment/{id}

Create External Booking

POST /api/shared-bookings

Approve External Booking

PUT /api/shared-bookings/{id}/approve

Reject External Booking

PUT /api/shared-bookings/{id}/reject

Partnership Management

POST /api/partnerships

GET /api/partnerships

PUT /api/partnerships/{id}

DELETE /api/partnerships/{id}

Cost Calculation

26.14 Validation Rules

- Institutions must have an active partnership.
 - Agreement must not be expired.
 - Equipment must be marked as shareable.
 - Equipment must be available.
 - Booking dates must not overlap.
 - Security deposit must be non-negative.
 - Hourly and daily rates must be valid.
-

26.15 Business Rules

BR-061

Only institutions with an active partnership can access shared equipment.

BR-062

Only equipment marked as "Shareable" appears in the shared catalog.

BR-063

External bookings always require approval from the owning institution unless auto-approval is configured.

BR-064

Equipment under maintenance cannot be shared.

BR-065

Sharing costs are calculated based on the institution's configured pricing policy.

BR-066

Agreement expiration immediately disables new sharing requests.

BR-067

Availability changes are synchronized in real time across partner institutions.

BR-068

Damage reports must be recorded before closing external bookings if applicable.

BR-069

Every external transaction generates an audit log.

BR-070

Institution Administrators can suspend sharing at any time.

26.16 Exception Handling

Exception	HTTP Status
Partnership Not Found	404
Partnership Expired	403
Equipment Not Shareable	400
External Booking Conflict	409
Unauthorized	401
Forbidden	403
Cost Calculation Failed	500

26.17 Acceptance Criteria

ID	Requirement
AC-071	Partner institutions can search shared equipment
AC-072	External booking requests can be submitted
AC-073	Approval workflow functions correctly
AC-074	Availability synchronizes across institutions
AC-075	Sharing charges are calculated accurately
AC-076	Partnership rules are enforced
AC-077	External booking history is maintained
AC-078	Notifications are sent on approval/rejection
AC-079	Audit logs record all sharing activities
AC-080	Reports include inter-institution usage statistics

26.18 Analytics

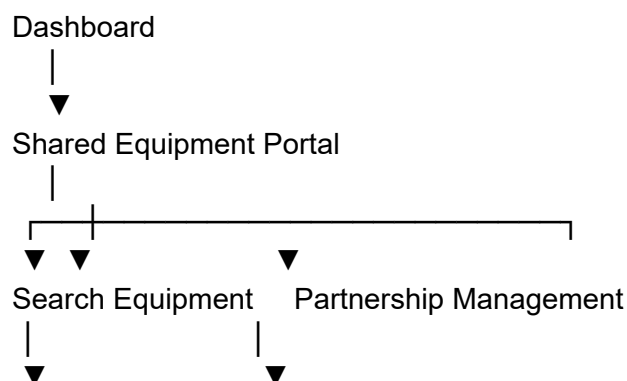
Resource Sharing Dashboard displays:

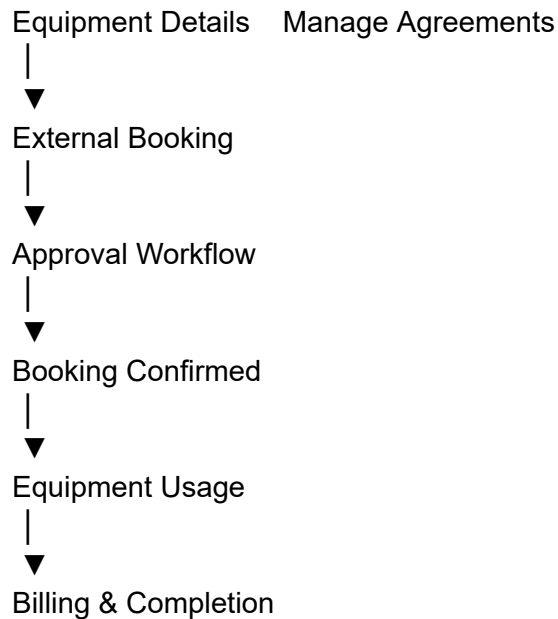
- Total Partner Institutions
 - Active Sharing Agreements
 - Shared Equipment Count
 - External Bookings
 - Revenue Generated
 - Cost Savings
 - Top Shared Equipment
 - Institution Collaboration Index
 - Average Approval Time
 - External Utilization Percentage
-

26.19 Future Enhancements

- National research equipment exchange network
 - AI-based institution matching
 - Blockchain-backed sharing agreements
 - Smart contract-based billing
 - Cross-border collaboration support
 - Automated insurance verification
 - Federated identity (SAML/OIDC)
 - Shared equipment recommendation engine
 - Carbon savings dashboard through shared utilization
 - Government-funded equipment sharing integration
-

26.20 Screen Flow





✓ End of Part 9

The PRD now defines a complete **multi-institution collaboration framework**, including partnership management, external bookings, cost allocation, synchronization, and governance.

Next (Part 10): Notification & Communication Management

This module will cover:

- Email, SMS, and in-app notifications
- Notification templates
- Event-driven notification engine
- Reminder scheduling
- Escalation rules
- User notification preferences
- Messaging APIs
- Delivery tracking
- Business rules and acceptance criteria

This module will ensure that users stay informed at every stage of the platform's workflows.

27. Module 8 – Notification & Communication Management

27.1 Module Overview

The Notification & Communication Management module provides a centralized event-driven communication system for the Lab Resource Utilization Platform. It delivers real-time notifications, reminders, alerts, and announcements through multiple channels, including in-app notifications, email, SMS, and push notifications (future enhancement).

The module improves user engagement, reduces missed bookings, ensures timely maintenance, and supports operational transparency.

27.2 Module Objectives

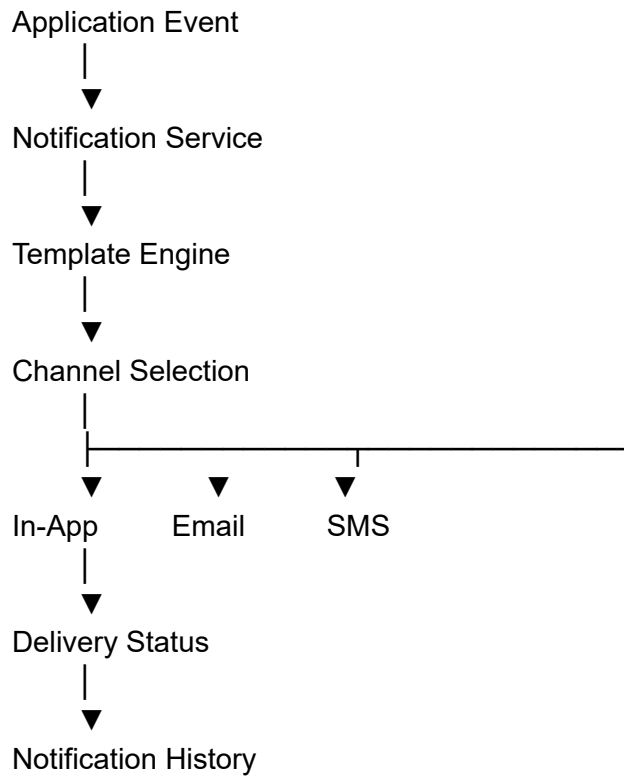
The Notification Module shall:

- Send real-time notifications.
 - Deliver booking reminders.
 - Notify users of approvals and rejections.
 - Alert technicians about maintenance tasks.
 - Send calibration reminders.
 - Broadcast institution-wide announcements.
 - Track notification delivery.
 - Allow users to manage notification preferences.
 - Retry failed notifications.
 - Maintain notification history.
-

27.3 Notification Channels

Channel	Description
In-App Notification	Displayed within the application
Email	Sent to registered email address
SMS	Sent to registered mobile number
Push Notification (<i>Future</i>)	Mobile application notifications
Webhook (<i>Future</i>)	Integration with external systems

27.4 Event-Driven Notification Flow



27.5 Notification Events

Authentication

- User Registration Successful
- Email Verification
- Password Reset
- Login from New Device
- Account Locked
- Password Changed

Equipment

- New Equipment Added
- Equipment Updated
- Equipment Retired
- Equipment Status Changed

Booking

- Booking Created
 - Booking Approved
 - Booking Rejected
 - Booking Cancelled
 - Booking Reminder (24 Hours)
 - Booking Reminder (1 Hour)
 - Equipment Ready
 - Booking Completed
 - Waitlist Promoted
-

Maintenance

- Work Order Created
 - Technician Assigned
 - Maintenance Started
 - Maintenance Completed
 - Calibration Due
 - Calibration Overdue
-

Institution

- Partnership Invitation
 - Partnership Approved
 - New Announcement
 - Policy Updated
-

27.6 Notification Templates

Each notification template contains:

- Subject
- Title
- Message
- Priority
- Channel
- Language
- Variables
- Status

Example:

Booking Approved

Subject:

Booking Approved

Message:

Your booking for {{equipmentName}} has been approved.

Booking Date:

{{bookingDate}}

Time:

{{startTime}} - {{endTime}}

27.7 User Notification Preferences

Users can configure:

- Enable Email
 - Enable SMS
 - Enable In-App
 - Enable Push Notifications
 - Reminder Frequency
 - Quiet Hours
 - Language Preference
-

27.8 Notification Priority

Priority	Description
Critical	Immediate delivery
High	Delivered instantly
Medium	Delivered within a few minutes
Low	Batched delivery

27.9 Screens

NT-01 – Notification Center

Displays:

- All notifications
 - Read / Unread status
 - Filters
 - Search
 - Delete
 - Mark as Read
-

NT-02 – Notification Preferences

Allows users to configure:

- Email Notifications
 - SMS Notifications
 - In-App Notifications
 - Reminder Settings
 - Language
-

NT-03 – Announcement Center

Displays:

- Institution Announcements
 - Department Announcements
 - System Announcements
-

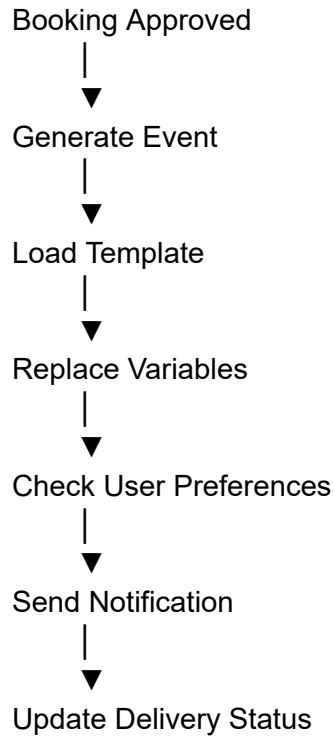
NT-04 – Notification History

Displays:

- Sent Time
- Delivery Status
- Channel
- Retry Count

- Open Status
-

27.10 Notification Workflow



27.11 Reminder Engine

Automatic reminders for:

- Upcoming bookings (24 hours before)
 - Upcoming bookings (1 hour before)
 - Maintenance due
 - Calibration due
 - Partnership expiry
 - Warranty expiry
 - Annual compliance checks
-

27.12 Announcement Management

Institution Administrators can create announcements with:

- Title
- Description
- Audience
- Start Date
- End Date
- Priority
- Attachments

Target audiences:

- All Users
- Institution
- Department
- Laboratory
- Specific Roles

27.13 Database Tables

notifications

Column	Type
id	BIGINT
user_id	BIGINT
title	VARCHAR
message	TEXT
notification_type	VARCHAR
priority	VARCHAR
status	VARCHAR
created_at	TIMESTAMP

notification_templates

Column	Type
id	BIGINT

template_name	VARCHAR
subject	VARCHAR
body	TEXT
channel	VARCHAR
language	VARCHAR

notification_preferences

Column	Type
id	BIGINT
user_id	BIGINT
email_enabled	BOOLEAN
sms_enabled	BOOLEAN
in_app_enabled	BOOLEAN
language	VARCHAR

announcements

Column	Type
id	BIGINT
title	VARCHAR
description	TEXT
audience	VARCHAR
start_date	DATE
end_date	DATE
created_by	BIGINT

27.14 REST APIs

Get Notifications

GET /api/notifications

Mark Notification as Read

PUT /api/notifications/{id}/read

Delete Notification

DELETE /api/notifications/{id}

Update Preferences

PUT /api/notifications/preferences

Create Announcement

POST /api/announcements

Get Announcements

GET /api/announcements

27.15 Validation Rules

- Notification title is mandatory.
 - Message body cannot be empty.
 - Announcement end date must be after start date.
 - User preferences must exist before sending notifications.
 - Only authorized users can create announcements.
 - Templates must support required variables.
-

27.16 Business Rules

BR-071

Notifications must respect user communication preferences.

BR-072

Critical notifications cannot be disabled by users.

BR-073

Duplicate notifications for the same event should be prevented.

BR-074

Failed deliveries must be retried based on configurable retry policies.

BR-075

Expired announcements are automatically archived.

BR-076

Unread notifications remain available until deleted or archived.

BR-077

Notification templates support localization for multiple languages.

BR-078

Delivery status is recorded for every notification channel.

BR-079

Only Institution Administrators and System Administrators can publish announcements.

BR-080

Notification history must be retained for audit and troubleshooting.

27.17 Exception Handling

Exception	HTTP Status
Notification Not Found	404
Invalid Template	400
Invalid User Preference	400
Delivery Failure	500
Unauthorized	401
Forbidden	403

27.18 Acceptance Criteria

ID	Requirement
AC-081	Notifications are generated for supported events
AC-082	User preferences are respected
AC-083	Booking reminders are delivered on schedule
AC-084	Maintenance alerts reach assigned technicians
AC-085	Announcements are visible to the intended audience
AC-086	Notification history is available
AC-087	Failed notifications are retried
AC-088	Delivery status is tracked
AC-089	Critical alerts bypass optional notification settings
AC-090	Audit logs capture notification events

27.19 Notification Analytics

The Notification Dashboard displays:

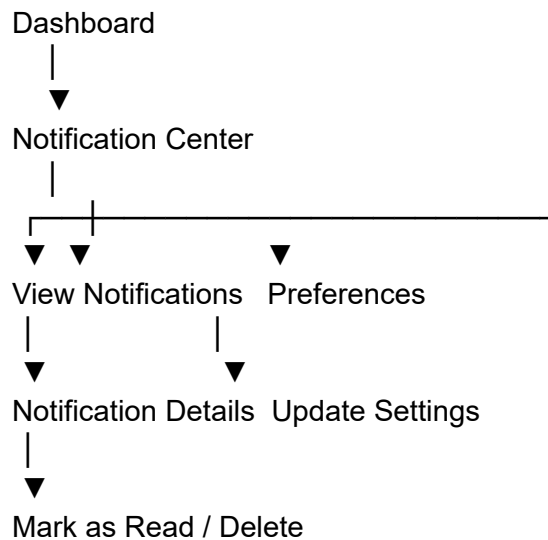
- Notifications Sent
 - Delivery Success Rate
 - Email Delivery Rate
 - SMS Delivery Rate
 - In-App Read Rate
 - Failed Deliveries
 - Average Delivery Time
 - Most Common Notification Types
 - User Engagement Rate
 - Announcement Reach
-

27.20 Future Enhancements

- Mobile push notifications

- Microsoft Teams integration
 - Slack integration
 - WhatsApp Business notifications
 - AI-powered notification scheduling
 - Rich media notifications (images/videos)
 - Voice call alerts for critical events
 - Read receipt analytics
 - Smart notification grouping
 - Notification fatigue detection
-

27.21 Screen Flow



✓ End of Part 10

The PRD now includes a comprehensive **Notification & Communication Management** module with event-driven architecture, templates, user preferences, reminders, announcements, APIs, business rules, and analytics.

Next (Part 11): Reporting, Billing & Audit Management

The next module will consolidate operational reporting, billing for shared resources, financial tracking, and audit logging. It will cover:

- Report generation and exports (PDF, Excel, CSV)
- Billing and invoicing for inter-institution sharing
- Payment tracking
- Audit logs and compliance
- Financial dashboards

- Scheduled reports
- Database schema
- APIs
- Business rules
- Acceptance criteria

This module completes the governance and compliance capabilities of the platform.

28. Module 9 – Reporting, Billing & Audit Management

28.1 Module Overview

The Reporting, Billing & Audit Management module provides comprehensive operational reporting, financial management for shared resource usage, and immutable audit logging for all significant system activities.

It enables administrators to generate reports, calculate usage-based charges, issue invoices, track payments, monitor compliance, and maintain detailed audit trails for security and governance.

This module integrates with:

- Booking Management
 - Resource Sharing
 - Maintenance
 - Analytics
 - Notification
 - Authentication
-

28.2 Module Objectives

The module shall:

- Generate operational reports.
- Generate financial reports.
- Calculate equipment usage charges.
- Generate invoices.
- Track payments.

- Export reports.
 - Maintain audit logs.
 - Support compliance audits.
 - Schedule recurring reports.
 - Provide financial dashboards.
-

28.3 Reporting Categories

Operational Reports

- Equipment Inventory Report
 - Equipment Utilization Report
 - Booking Report
 - Booking Cancellation Report
 - Booking Rejection Report
 - Maintenance Report
 - Calibration Report
 - Technician Performance Report
-

Administrative Reports

- Institution Summary
 - Department Summary
 - Laboratory Summary
 - User Activity Report
 - Login Activity Report
 - Role Assignment Report
-

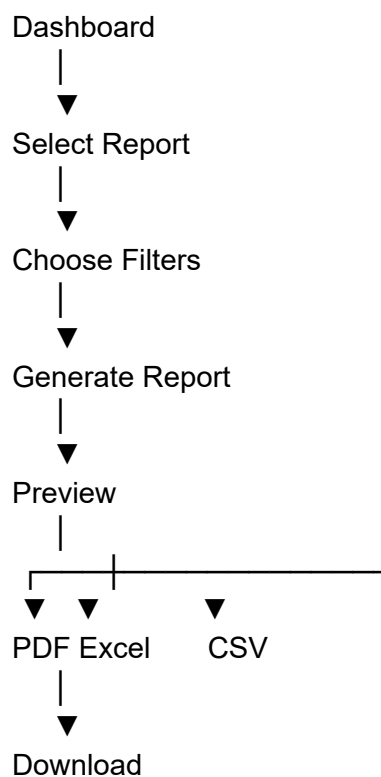
Financial Reports

- Equipment Usage Charges
 - Revenue Report
 - Outstanding Payments
 - Invoice Report
 - Institution Billing Summary
 - Resource Sharing Revenue
-

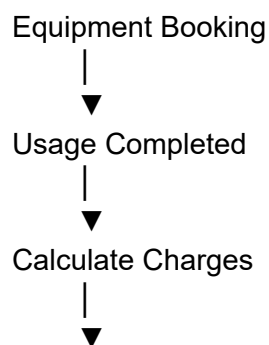
Compliance Reports

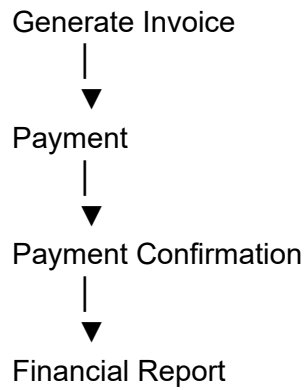
- Audit Trail
 - Data Access Report
 - Security Events
 - Configuration Changes
 - User Permission Changes
-

28.4 Reporting Workflow



28.5 Billing Workflow





28.6 Pricing Models

The platform supports:

- Hourly Pricing
 - Daily Pricing
 - Weekly Pricing
 - Monthly Pricing
 - Flat Rate
 - Security Deposit
 - Penalty Charges
 - Damage Charges
-

28.7 Invoice Components

Each invoice contains:

- Invoice Number
- Institution Name
- Equipment Details
- Booking ID
- Usage Duration
- Unit Price
- Taxes
- Discounts
- Security Deposit
- Total Amount
- Payment Status
- Invoice Date
- Due Date

28.8 Payment Status

Status	Description
Pending	Invoice generated
Paid	Payment received
Partially Paid	Partial payment received
Failed	Payment unsuccessful
Refunded	Payment refunded
Cancelled	Invoice cancelled

28.9 Audit Logging

Every critical operation generates an audit log.

Examples:

- User Login
 - User Logout
 - User Creation
 - Equipment Added
 - Equipment Deleted
 - Booking Approved
 - Booking Rejected
 - Maintenance Updated
 - Invoice Generated
 - Payment Recorded
 - Permission Changed
-

28.10 Screens

RP-01 – Report Dashboard

Displays:

- Available Reports
 - Recent Reports
 - Scheduled Reports
 - Favorites
-

RP-02 – Report Generator

Fields:

- Report Type
- Institution
- Department
- Laboratory
- Equipment
- Date Range
- Output Format

Buttons:

- Preview
 - Generate
 - Export
-

RP-03 – Invoice Dashboard

Displays:

- Pending Invoices
 - Paid Invoices
 - Overdue Invoices
 - Total Revenue
-

RP-04 – Invoice Details

Displays:

- Invoice Information
- Usage Details
- Payment History

- [Download PDF](#)
-

RP-05 – Audit Log Viewer

Displays:

- Timestamp
- User
- Action
- Module
- IP Address
- Result
- Details

Supports:

- Search
 - Filter
 - Export
-

28.11 Scheduled Reports

Administrators can schedule reports:

- Daily
- Weekly
- Monthly
- Quarterly
- Yearly

Delivery Channels:

- Email
 - In-App Notification
 - Download Center
-

28.12 Database Tables

reports

Column	Type
id	BIGINT
report_name	VARCHAR
report_type	VARCHAR
generated_by	BIGINT
generated_at	TIMESTAMP
file_path	VARCHAR

invoices

Column	Type
id	BIGINT
invoice_number	VARCHAR
institution_id	BIGINT
booking_id	BIGINT
total_amount	DECIMAL
payment_status	VARCHAR
due_date	DATE
generated_at	TIMESTAMP

payments

Column	Type
id	BIGINT
invoice_id	BIGINT
payment_reference	VARCHAR
amount_paid	DECIMAL

payment_method	VARCHAR
payment_date	TIMESTAMP
payment_status	VARCHAR

audit_logs

Column	Type
id	BIGINT
user_id	BIGINT
module	VARCHAR
action	VARCHAR
entity_type	VARCHAR
entity_id	BIGINT
ip_address	VARCHAR
action_time	TIMESTAMP
result	VARCHAR

28.13 REST APIs

Generate Report

POST /api/reports/generate

Download Report

GET /api/reports/{id}/download

List Reports

GET /api/reports

Generate Invoice

POST /api/invoices

Get Invoice

GET /api/invoices/{id}

Record Payment

POST /api/payments

Get Audit Logs

GET /api/audit-logs

Export Audit Logs

GET /api/audit-logs/export

28.14 Validation Rules

- Report type is mandatory.
 - Date range must be valid.
 - Invoice number must be unique.
 - Payment amount cannot exceed outstanding balance.
 - Due date must not precede invoice date.
 - Only authorized roles can access audit logs.
 - Export format must be supported.
-

28.15 Business Rules

BR-081

Every completed shared booking generates an invoice if billing is enabled.

BR-082

Invoice numbers are generated automatically and must be unique.

BR-083

Audit logs cannot be modified or deleted by application users.

BR-084

Scheduled reports run according to the configured schedule.

BR-085

Financial reports include only finalized transactions.

BR-086

Payment status updates automatically after successful payment confirmation.

BR-087

Audit logs record both successful and failed security-sensitive operations.

BR-088

Reports respect role-based data access restrictions.

BR-089

Invoice totals include taxes, discounts, and additional charges as configured.

BR-090

Generated reports are retained according to the organization's data retention policy.

28.16 Exception Handling

Exception	HTTP Status
Report Not Found	404
Invalid Report Parameters	400
Invoice Not Found	404
Payment Failed	500
Unauthorized	401
Forbidden	403
Export Failed	500

28.17 Acceptance Criteria

ID	Requirement
AC-091	Reports generate successfully in supported formats

- AC-092 Scheduled reports execute automatically
- AC-093 Invoices are generated accurately
- AC-094 Payments are recorded correctly
- AC-095 Financial dashboards reflect current data
- AC-096 Audit logs capture all configured events
- AC-097 Audit logs are searchable and exportable
- AC-098 Reports enforce role-based visibility
- AC-099 Invoice calculations match configured pricing rules
- AC-100 Compliance reports are available for audit purposes
-

28.18 Financial Dashboard

Displays:

- Total Revenue
 - Revenue by Institution
 - Outstanding Payments
 - Paid Invoices
 - Average Invoice Value
 - Monthly Revenue Trend
 - Top Revenue-Generating Equipment
 - Security Deposit Balance
 - Refund Summary
 - Collection Rate
-

28.19 Compliance Dashboard

Displays:

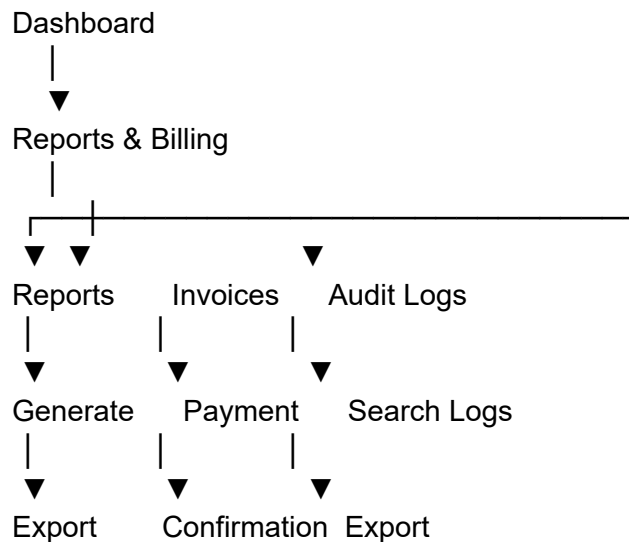
- Total Audit Events
- Failed Login Attempts
- Permission Changes
- Configuration Changes
- Sensitive Data Access
- Export Activities

- Security Alerts
 - Compliance Status
-

28.20 Future Enhancements

- Integration with payment gateways (Stripe, Razorpay, PayPal)
 - GST/VAT automation
 - ERP integration (SAP, Oracle ERP)
 - Digital invoice signing
 - AI-powered financial forecasting
 - Automated reconciliation
 - Fraud detection analytics
 - Immutable blockchain audit ledger
 - Electronic compliance submissions
 - Executive financial reporting portal
-

28.21 Screen Flow



✓ End of Part 11

At this point, the PRD covers all major functional modules:

- Authentication & Authorization
- Institution Management
- Equipment Inventory

- Booking & Scheduling
- Real-Time Analytics
- Maintenance & Calibration
- Inter-Institution Resource Sharing
- Notification & Communication
- **Reporting, Billing & Audit Management**

Next (Part 12 – Final Enterprise Sections)

The final part will include the cross-cutting sections expected in a production-ready PRD:

- Non-Functional Requirements (Performance, Security, Scalability, Availability)
- System Architecture Overview
- Integration Requirements
- Deployment Strategy
- Backup & Disaster Recovery
- Logging & Monitoring
- Security & Compliance
- Testing Strategy
- Risks & Mitigation
- Assumptions & Dependencies
- Release Plan & Milestones
- Appendix, Glossary, and Traceability Matrix

This will complete the PRD as a document suitable for handing over to architects, developers, QA engineers, and project stakeholders.

29. Non-Functional Requirements (NFR)

29.1 Overview

Non-functional requirements define **how** the system should perform rather than **what** it should do. These requirements ensure the platform is secure, scalable, reliable, maintainable, and performant.

29.2 Performance Requirements

Response Time

Operation	Maximum Response Time
Login	≤ 2 seconds

Equipment Search	≤ 2 seconds
Booking Creation	≤ 3 seconds
Dashboard Loading	≤ 4 seconds
Report Generation	≤ 30 seconds
QR Code Generation	≤ 5 seconds

Concurrent Users

The platform shall support:

- 1,000 concurrent users (initial deployment)
 - 10,000 registered users
 - 100 simultaneous booking requests
 - 500 concurrent API requests
-

Database Performance

- Indexed search fields
 - Query optimization
 - Connection pooling
 - Pagination for large datasets
-

29.3 Scalability

The system shall support horizontal scaling.

Architecture should support:

- Multiple Application Servers
 - Load Balancer
 - Distributed Cache
 - Cloud Storage
 - Read Replicas
 - Container Deployment
-

29.4 Availability

Requirement	Target
System Availability	99.9%
Scheduled Maintenance	Off-Peak Hours
Maximum Downtime	< 8 hours/year
Database Recovery	< 30 minutes

29.5 Reliability

The platform shall:

- Recover from server failures automatically.
 - Prevent data loss.
 - Support transactional consistency.
 - Detect service failures.
-

29.6 Security Requirements

Authentication

- JWT Authentication
- OAuth2 Login
- Password Encryption (BCrypt)

Authorization

- Role-Based Access Control (RBAC)
- Resource-Level Permissions
- API Authorization

Data Security

- HTTPS Everywhere
- TLS 1.3
- Password Hashing
- Secure Cookies
- CSRF Protection

- XSS Protection
- SQL Injection Prevention

Audit

- Login Logs
 - Activity Logs
 - Permission Changes
 - Failed Login Attempts
-

29.7 Privacy

The system shall:

- Store only necessary personal information.
 - Encrypt sensitive information at rest.
 - Encrypt all communication in transit.
 - Allow configurable data retention policies.
 - Support anonymization for archived analytics where required.
-

29.8 Maintainability

The project shall follow:

- Clean Architecture
 - SOLID Principles
 - Layered Architecture
 - Dependency Injection
 - Modular Development
 - Code Documentation
-

29.9 Usability

The UI should be:

- Responsive
- Mobile Friendly
- Accessible

- Easy Navigation
 - Consistent Design
 - Minimal Clicks
-

29.10 Compatibility

Supported Browsers

- Chrome
- Edge
- Firefox
- Safari

Supported Devices

- Desktop
 - Tablet
 - Mobile Browser
-

29.11 Logging

The application shall log:

- API Requests
 - Errors
 - Exceptions
 - User Activities
 - Authentication Events
 - Database Errors
-

29.12 Monitoring

Monitor:

- CPU Usage
- Memory Usage
- Database Health
- API Latency

- Error Rates
 - Storage Usage
-

29.13 Backup

Backup Strategy

Database

- Daily Incremental Backup
- Weekly Full Backup
- Monthly Archive

Files

- Daily Backup
 - Versioning Enabled
-

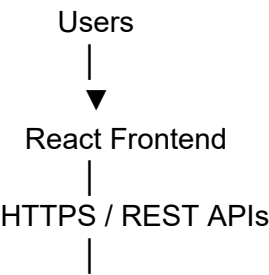
29.14 Disaster Recovery

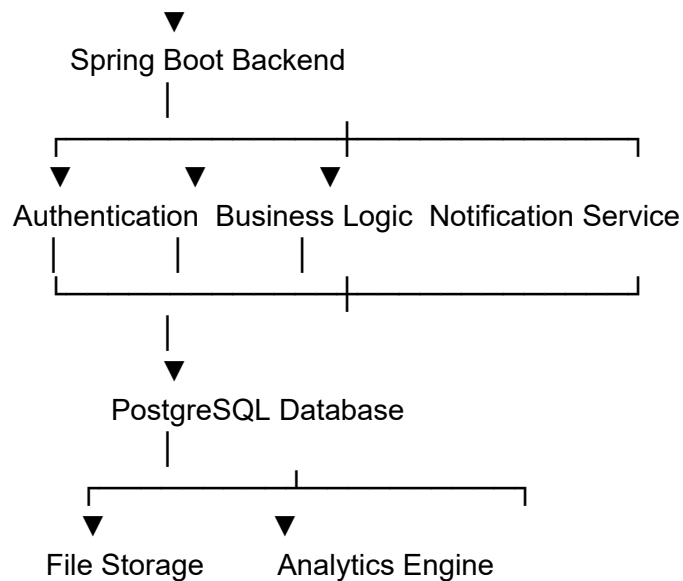
Recovery Objectives

Metric	Target
RPO (Recovery Point Objective)	15 minutes
RTO (Recovery Time Objective)	30 minutes

30. System Architecture

30.1 High-Level Architecture





30.2 Technology Stack

Frontend

- React.js
- React Router
- Axios
- Material UI (or Bootstrap)
- Chart.js

Backend

- Spring Boot
- Spring Security
- Spring Data JPA
- Hibernate
- JWT
- OAuth2

Database

- PostgreSQL

Storage

- Local Storage (Development)
 - AWS S3 / Azure Blob Storage (Production)
-

Build Tools

- Maven
 - npm
-

Version Control

- Git
 - GitHub
-

31. Integration Requirements

External integrations may include:

- Email Service (SMTP)
 - SMS Gateway
 - QR Code Generator
 - Payment Gateway
 - Cloud Storage
 - Identity Provider (OAuth2)
 - ERP Systems (Future)
 - Calendar Services (Future)
-

32. Deployment Strategy

Development Environment

- Local Machine
- Docker (Optional)

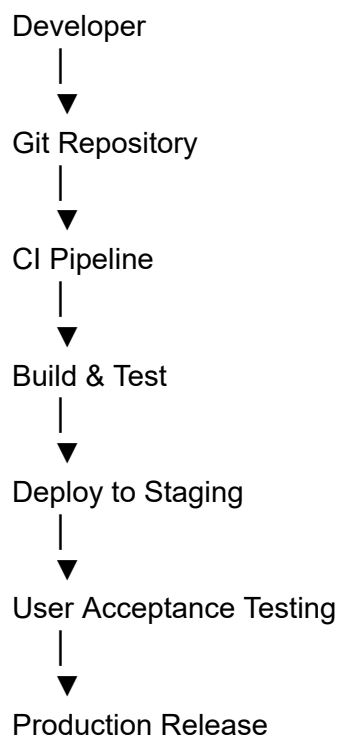
Testing Environment

- QA Server
 - Staging Database
-

Production Environment

- Linux Server
 - Nginx Reverse Proxy
 - Spring Boot Service
 - PostgreSQL Database
-

Deployment Flow



33. Testing Strategy

Test Levels

- Unit Testing
 - Integration Testing
 - System Testing
 - Regression Testing
 - User Acceptance Testing (UAT)
 - Performance Testing
 - Security Testing
-

Test Coverage

Target:

- Backend Unit Tests $\geq 80\%$
 - Critical Business Logic $\geq 90\%$
-

34. Risk Analysis

Risk	Impact	Mitigation
Server Failure	High	Redundant deployment
Database Failure	High	Replication & backups
Security Breach	High	Encryption, RBAC, monitoring
High Traffic	Medium	Load balancing
Third-party Service Failure	Medium	Retry mechanisms & graceful degradation
Data Corruption	High	Transaction management & backups

35. Assumptions

- All institutions have reliable internet access.
 - Equipment data is maintained accurately.
 - Users receive required training.
 - Authentication services are available.
 - Partner institutions comply with sharing agreements.
-

36. Dependencies

- React Frontend
 - Spring Boot Backend
 - PostgreSQL Database
 - Email Service
 - File Storage
 - Notification Service
 - QR Code Library
 - Payment Gateway (Optional)
-

37. Release Plan

Phase 1

- Authentication
 - Institution Management
 - Equipment Inventory
-

Phase 2

- Booking
 - Maintenance
 - Notifications
-

Phase 3

- Analytics
 - Resource Sharing
 - Billing
-

Phase 4

- AI Features
- Mobile App

- ERP Integrations
-

38. Success Metrics

The project will be considered successful if it achieves:

KPI	Target
Equipment Utilization	> 80%
Booking Success Rate	> 95%
Average Booking Response Time	< 3 seconds
User Satisfaction Score	> 4.5 / 5
System Availability	≥ 99.9%
Preventive Maintenance Compliance	> 95%
Report Generation Success	> 99%
Notification Delivery Success	> 98%

39. Glossary

Term	Meaning
Equipment	Laboratory asset managed by the system
Booking	Reservation of equipment for a defined time slot
Calibration	Verification of measurement accuracy
Work Order	Maintenance task assigned to a technician
Institution	University or research organization using the platform
Department	Academic or research division within an institution
Laboratory	Physical location where equipment is housed

Utilization	Percentage of available time during which equipment is actively used
RBAC	Role-Based Access Control
KPI	Key Performance Indicator

40. Requirements Traceability Matrix (RTM)

Business Goal	Module	Requirement IDs
Secure access	Authentication	AC-001 to AC-010
Organizational management	Institution Management	AC-021 to AC-030
Asset management	Equipment Inventory	AC-031 to AC-040
Equipment scheduling	Booking	AC-041 to AC-050
Operational insights	Analytics	AC-051 to AC-060
Equipment reliability	Maintenance	AC-061 to AC-070
Collaboration	Resource Sharing	AC-071 to AC-080
User communication	Notifications	AC-081 to AC-090
Governance & compliance	Reporting/Billing/Audit	AC-091 to AC-100

41. Appendix

The appendix may include:

- Wireframes
- UI mockups
- ER diagrams
- API specifications (OpenAPI/Swagger)
- Database schema scripts
- Sample reports
- User manuals
- Deployment guides
- Configuration guides

- Test cases
-

42. Document Sign-Off

Role	Responsibility
Product Manager	Approves business requirements
Solution Architect	Validates technical feasibility
UI/UX Designer	Confirms design alignment
Development Lead	Reviews implementation readiness
QA Lead	Reviews testability
Project Manager	Approves project scope
Client/Stakeholder	Final business approval



End of Enterprise PRD

This completes a comprehensive enterprise-style Product Requirements Document. It includes:

- **42 structured sections**
- **9 major functional modules**
- **100 acceptance criteria**
- **90 business rules**
- **REST API definitions**
- **Database schema**
- **Screen inventories**
- **Workflows and process flows**
- **Architecture and deployment guidance**
- **Testing, security, reporting, governance, and operational requirements**

This document is suitable as a foundation for design, development, QA planning, and project execution, while leaving room for detailed UI designs, OpenAPI specifications, and low-level technical design documents that are typically produced in subsequent project phases.

continue with **Part 2** of the PRD.

This section is much closer to what product companies (Infosys, Microsoft, Atlassian, Zoho, etc.) write before development starts.

13. Target Users

The Lab Resource Utilization Platform is designed for multiple categories of users within universities, research institutions, laboratories, and collaborating organizations. Each user type has distinct responsibilities and system permissions.

13.1 Primary Users

1. Research Student

Description

Research students use laboratory equipment for academic research, experimentation, and project work.

Responsibilities

- Search laboratory equipment.
- View equipment availability.
- Request equipment bookings.
- View booking history.
- Receive booking notifications.
- Cancel bookings when required.

Goals

- Quickly locate available equipment.
 - Book equipment without manual approvals where applicable.
 - Receive reminders before scheduled usage.
 - Track previous equipment usage.
-

2. Faculty Member

Description

Faculty members supervise students and frequently reserve laboratory equipment for teaching and research activities.

Responsibilities

- Reserve equipment
- Approve student requests (department policy dependent)
- View departmental equipment usage
- Monitor research resource allocation

Goals

- Efficient scheduling
 - Better research planning
 - Reduced booking conflicts
-

3. Lab Technician

Description

Responsible for equipment maintenance and operational readiness.

Responsibilities

- Update equipment status
- Perform maintenance
- Record calibration details
- Upload maintenance reports
- Close maintenance requests

Goals

- Ensure equipment availability
 - Reduce downtime
 - Maintain maintenance records
-

4. Lab Manager

Description

Manages the complete laboratory and all equipment under a department.

Responsibilities

- Add new equipment
- Approve bookings
- Assign technicians
- View utilization reports
- Manage maintenance schedules

Goals

- Improve equipment utilization
 - Reduce idle equipment
 - Improve scheduling efficiency
-

5. Department Head

Description

Responsible for monitoring laboratory performance across departments.

Responsibilities

- Review reports
 - Monitor equipment utilization
 - Approve procurement recommendations
 - View department analytics
-

6. Institution Administrator

Description

Responsible for managing all laboratories within an institution.

Responsibilities

- Manage users
 - Manage departments
 - Configure institution
 - Generate organization reports
 - Monitor equipment inventory
-

7. System Administrator

Description

Platform owner responsible for overall application administration.

Responsibilities

- Manage institutions
- Configure platform
- Monitor system health

- Audit logs
 - Security management
-

14. Secondary Users

Secondary users interact indirectly with the platform.

Finance Department

Responsibilities

- Generate invoices
 - Track usage costs
 - Budget allocation
 - Department billing
-

Maintenance Team

Responsibilities

- Work orders
 - Spare part management
 - Maintenance planning
-

External Research Institutions

Responsibilities

- Request shared equipment
 - Manage external bookings
 - Track resource sharing
-

15. User Persona

Persona 1

Name

Arun Kumar

Role

Research Student

Age

23

Department

Mechanical Engineering

Goals

- Book CNC Machine
- Complete Final Year Project
- Receive booking reminders

Pain Points

- Doesn't know machine availability
- Manual booking takes time
- No booking history

Solution

Online booking platform

Persona 2

Name

Dr. Priya

Role

Faculty Member

Goals

- Reserve lab for practical sessions

- Approve student bookings

Pain Points

- Booking conflicts
- Manual approval

Solution

Department dashboard

Persona 3

Name

Rajesh

Role

Lab Technician

Goals

- Update maintenance
- Schedule calibration

Pain Points

- Paper maintenance logs

Solution

Maintenance dashboard

Persona 4

Name

Karthik

Role

Institution Admin

Goals

- Improve utilization
- Reduce procurement cost

Pain Points

- No centralized reports

Solution

Analytics Dashboard

16. Role & Permission Matrix

Feature	Student	Faculty	Technician	Manager	Dept Head	Institution Admin	System Admin
Login	✓	✓	✓	✓	✓	✓	✓
Register	✓	✓	✗	✗	✗	✗	✗
Search Equipment	✓	✓	✓	✓	✓	✓	✓
View Equipment	✓	✓	✓	✓	✓	✓	✓
Book Equipment	✓	✓	✗	✓	✗	✗	✗
Cancel Booking	✓	✓	✗	✓	✗	✗	✗
Approve Booking	✗	Optional	✗	✓	✓	✓	✓
Add Equipment	✗	✗	✗	✓	✗	✓	✓
Edit Equipment	✗	✗	✓	✓	✗	✓	✓
Delete Equipment	✗	✗	✗	✓	✗	✓	✓

Manage Users	×	×	×	×	×	✓	✓
Manage Departments	×	×	×	×	×	✓	✓
Manage Institutions	×	×	×	×	×	×	✓
View Reports	Limited	Department	Maintenance	Department	Department	Institution	Global
Generate Reports	×	×	×	✓	✓	✓	✓
Configure System	×	×	×	×	×	Limited	✓

17. User Stories

Authentication

US-001

As a user,

I want to log in securely

So that I can access my dashboard.

Acceptance Criteria

- Valid email required.
 - Valid password required.
 - JWT token generated.
 - Invalid credentials show error.
 - Redirect to dashboard after login.
-

Equipment Search

US-002

As a researcher,

I want to search laboratory equipment

So that I can locate available resources.

Acceptance Criteria

- Search by equipment name.
 - Search by category.
 - Search by department.
 - Search by institution.
 - Show equipment availability.
-

Booking Equipment

US-003

As a researcher,

I want to reserve equipment

So that I can perform my experiment.

Acceptance Criteria

- Equipment must be available.
 - Booking time should not overlap.
 - Confirmation notification sent.
 - Booking stored in database.
-

Equipment Approval

US-004

As a Lab Manager,

I want to approve bookings

So that equipment usage is controlled.

Acceptance Criteria

- View pending bookings.
 - Approve or reject.
 - Notify researcher.
 - Update booking status.
-

Maintenance

US-005

As a Technician,

I want to record maintenance

So that equipment history is maintained.

Acceptance Criteria

- Equipment status changes.
 - Maintenance log created.
 - Completion date stored.
 - Equipment becomes available after maintenance.
-

Analytics

US-006

As an Institution Admin,

I want utilization reports

So that procurement decisions can be made.

Acceptance Criteria

- Daily reports.
 - Monthly reports.
 - Equipment utilization.
 - Department comparison.
 - Export PDF.
-

18. Business Rules

The following business rules govern the application.

BR-001

A user can belong to only one institution.

BR-002

Every department belongs to one institution.

BR-003

Every laboratory belongs to one department.

BR-004

Equipment belongs to one laboratory.

BR-005

Equipment cannot be booked while under maintenance.

BR-006

Booking dates cannot overlap.

BR-007

Only managers can approve bookings.

BR-008

Only technicians can close maintenance tasks.

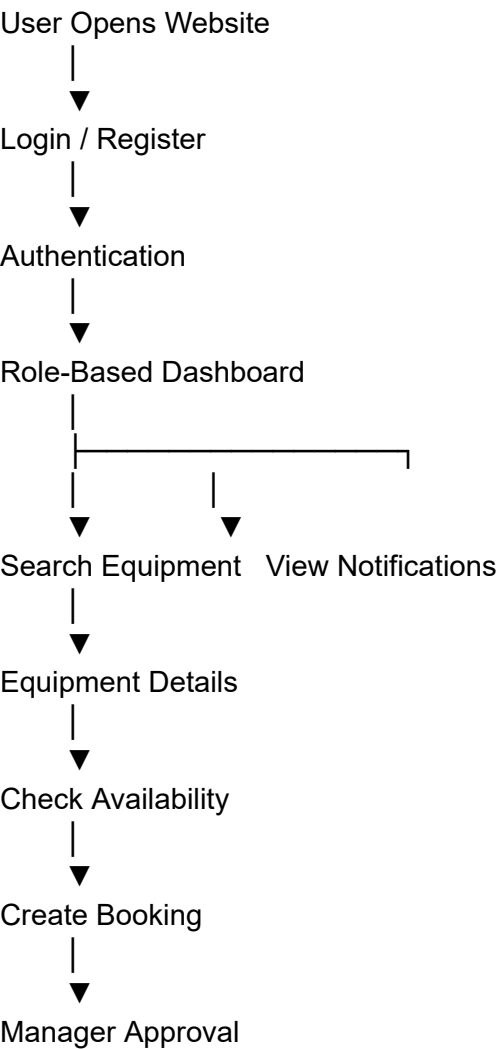
BR-009

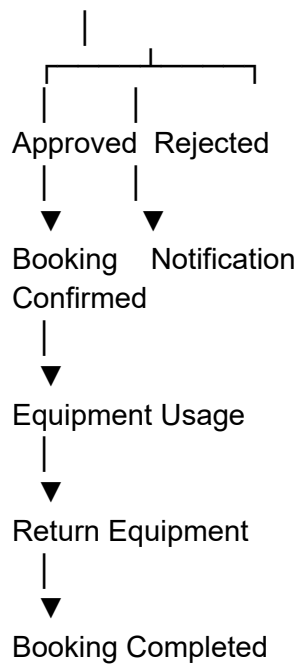
Deleted equipment cannot be booked.

BR-010

Every booking generates an audit log.

19. Functional Flow (High-Level User Journey)





End of Part 2

At this stage, the PRD has reached the level where a **UI/UX designer and backend architect** can begin planning. The next part will dive into **Module 1 (Authentication & Authorization)** with complete workflows, validations, database tables, APIs, screen details, sequence diagrams, and edge cases—essentially documenting it the way an enterprise team would before implementation.

20. Module 1 – Authentication & Authorization

20.1 Module Overview

Authentication is the first and most critical module of the Lab Resource Utilization Platform. It ensures that only authorized users can access the system while enforcing role-based permissions.

This module provides secure login, registration, JWT-based session management, OAuth2 login, password recovery, and user profile management.

The module is built using **Spring Security**, **JWT**, **OAuth2**, **BCrypt Password Encoder**, and **React Authentication Context**.

20.2 Objectives

The Authentication module should:

- Secure user credentials.
 - Authenticate users using JWT.
 - Authorize users based on assigned roles.
 - Protect all APIs.
 - Support Google OAuth Login.
 - Support Password Reset.
 - Maintain user sessions securely.
 - Prevent unauthorized access.
 - Record authentication logs.
-

20.3 Functional Scope

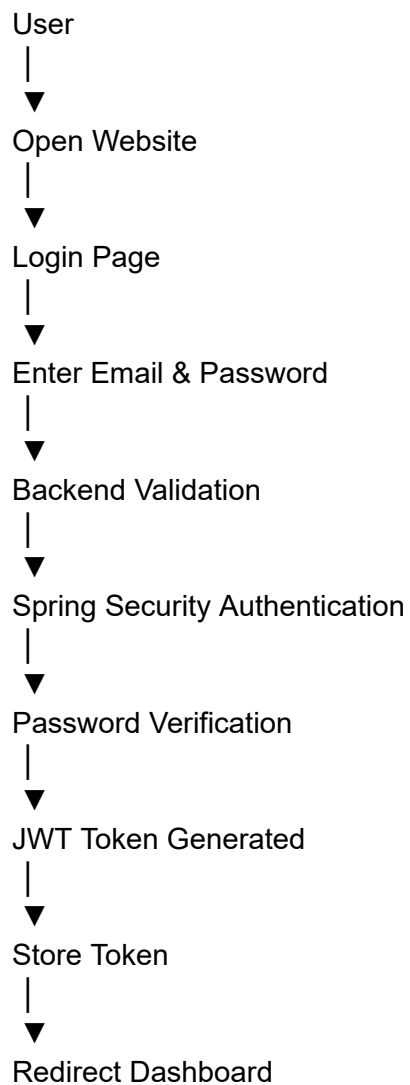
Included Features

- ✓ User Registration
 - ✓ Login
 - ✓ JWT Authentication
 - ✓ Google OAuth2 Login
 - ✓ Forgot Password
 - ✓ Reset Password
 - ✓ Refresh Access Token
 - ✓ Logout
 - ✓ Profile Management
 - ✓ Change Password
 - ✓ Role-Based Authorization
 - ✓ Session Validation
 - ✓ Authentication Logs
-

20.4 Actors

Actor	Access
Student	Register, Login
Faculty	Login
Lab Technician	Login
Lab Manager	Login
Institution Admin	Login
System Admin	Login

20.5 Authentication Workflow



20.6 Login Screen

Purpose

Allows registered users to access the application securely.

Components

Email

Password

Remember Me Checkbox

Forgot Password Link

Login Button

Google Login Button

Register Link

Validation Rules

Email required

Password required

Valid email format

Minimum password length

Maximum password length

Prevent SQL Injection

Prevent XSS

Rate Limiting

Success Flow

Login



Authentication Success



JWT Generated



Save Token



Dashboard

Failure Flow

Login



Invalid Credentials



Show Error



Retry Login

20.7 Registration Screen

Purpose

Allows new researchers to create accounts.

Fields

First Name

Last Name

Email

Phone Number

Institution

Department

Password

Confirm Password

Terms & Conditions

Validations

Email must be unique

Phone unique

Password minimum 8 characters

One uppercase

One lowercase

One number

One special character

Passwords must match

Institution mandatory

Department mandatory

Registration Flow

Fill Form

↓

Validate



Save User



Encrypt Password



Send Verification Email



Account Created

20.8 Forgot Password Flow

Forgot Password



Enter Email



Verify Email Exists



Generate Reset Token



Send Email



User Opens Link



Reset Password

↓

Login

20.9 Change Password Flow

Dashboard

↓

Profile

↓

Change Password

↓

Old Password

↓

New Password

↓

Confirm Password

↓

Update

↓

Logout

↓

Login Again

20.10 JWT Authentication Flow

Client



Login API



Spring Security



Authentication Manager



User Details Service



Validate Password



Generate JWT



Return Token



Client Stores Token



API Requests



Authorization Header



JWT Validation



20.11 JWT Token Structure

Header

Payload

Signature

Payload Example

```
{
  "userId":101,
  "email":"student@abc.edu",
  "role":"RESEARCHER",
  "institutionId":5,
  "exp":"2026-07-20T18:30:00"
}
```

20.12 Role-Based Authorization

Role	Dashboard
Student	Research Dashboard
Faculty	Faculty Dashboard
Technician	Maintenance Dashboard
Manager	Lab Dashboard
Institution Admin	Institution Dashboard
System Admin	Global Dashboard

20.13 Screen Flow

Landing Page



Login



JWT Authentication



Role Verification



Dashboard

20.14 UI Components

Login Card

Email Textbox

Password Textbox

Remember Me

Google Login Button

Forgot Password Link

Login Button

Error Message

Loading Spinner

Footer

20.15 Database Tables

users

Field

Type

id	BIGINT
first_name	VARCHAR
last_name	VARCHAR
email	VARCHAR
phone	VARCHAR
password	VARCHAR
institution_id	BIGINT
department_id	BIGINT
role_id	BIGINT
status	BOOLEAN
created_at	TIMESTAMP

roles

Field	Type
id	BIGINT
role_name	VARCHAR

refresh_tokens

Field	Type
id	BIGINT
user_id	BIGINT
token	TEXT
expiry_date	TIMESTAMP

password_reset_tokens

Field	Type
-------	------

id	BIGINT
user_id	BIGINT
token	VARCHAR
expiry	TIMESTAMP

20.16 REST APIs

Register

POST /api/auth/register

Request

```
{
  "firstName": "Selva",
  "lastName": "K",
  "email": "selva@gmail.com",
  "password": "Password@123"
}
```

Response

```
{
  "message": "Registration Successful"
}
```

Login

POST /api/auth/login

Request

```
{
  "email": "selva@gmail.com",
  "password": "Password@123"
}
```

Response

```
{
  "accessToken": "JWT_TOKEN",
}
```

```
"refreshToken":"REFRESH_TOKEN",  
"role":"LAB_MANAGER"  
}
```

Forgot Password

POST /api/auth/forgot-password

Reset Password

POST /api/auth/reset-password

Refresh Token

POST /api/auth/refresh

Logout

POST /api/auth/logout

20.17 Exception Handling

Error	HTTP Code
Invalid Credentials	401
User Not Found	404
Email Exists	409
Invalid Token	401
Expired Token	401
Validation Failed	400
Internal Error	500

20.18 Business Rules

- Email must be unique.
 - Phone number must be unique.
 - Password stored only in encrypted form (BCrypt).
 - JWT expires after configured duration (e.g., 1 hour).
 - Refresh token required after access token expiry.
 - Locked users cannot log in.
 - Inactive users cannot log in.
 - Only System Admin can assign Institution Admin roles.
 - Every login and logout action should be recorded in the audit log.
-

20.19 Acceptance Criteria

ID	Criteria
AC-001	User can register with valid details
AC-002	Duplicate email is rejected
AC-003	Password is encrypted before storage
AC-004	Valid users receive JWT and Refresh Token
AC-005	Invalid credentials return HTTP 401
AC-006	Protected APIs require a valid JWT
AC-007	Role-based access is enforced
AC-008	Password reset works using secure tokens
AC-009	User can update profile details
AC-010	Audit log records authentication events

20.20 Future Enhancements

- Multi-Factor Authentication (MFA)
- Biometric Authentication
- Single Sign-On (SSO)
- Microsoft Azure AD Login
- LDAP Integration
- Device Management
- Login History Dashboard

- Session Management Console
-

End of Part 3

This module is now documented in enough detail that both frontend and backend developers can begin implementation.

Next Part (Part 4)

We'll document **Module 2 – Institution Management**, including:

- Institution hierarchy (Institution → Department → Lab)
- Database schema
- CRUD operations
- Screen flows
- APIs
- Validation rules
- User interface details
- Business rules
- Acceptance criteria
- Sequence diagrams
- Edge cases

This level of detail is what turns the PRD into a practical development blueprint rather than just a high-level requirements document.

Excellent. Now we're entering the **core business module** of the project.

This module is the **heart of the application** because every other module (Booking, Maintenance, Analytics, Billing, Reports) depends on it.

A real Product Manager would spend **15–20 pages** documenting this module.

22. Module 3 – Equipment Inventory Management

22.1 Module Overview

The Equipment Inventory Management Module is responsible for managing all laboratory equipment across institutions. It provides a centralized repository where administrators can register, organize, categorize, search, update, and monitor equipment throughout its lifecycle.

Every equipment item acts as an organizational asset and is uniquely identified within the platform. The module supports tracking equipment specifications, ownership, operational status, maintenance history, calibration records, location, documentation, and availability.

The module is the foundation for:

- Equipment Booking
- Resource Sharing
- Maintenance
- Billing
- Analytics
- Reports

Without this module, none of the remaining modules can function.

22.2 Module Objectives

The Equipment Inventory Module shall:

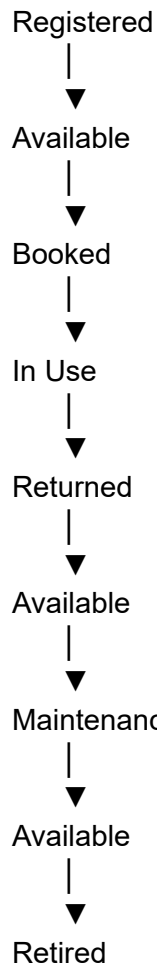
- Maintain a centralized inventory of all laboratory equipment.
 - Support equipment categorization.
 - Store equipment specifications and technical information.
 - Track equipment availability.
 - Track equipment ownership.
 - Maintain calibration records.
 - Store equipment manuals and documents.
 - Generate QR Codes.
 - Support advanced searching.
 - Maintain equipment lifecycle history.
-

22.3 Equipment Lifecycle

Every equipment passes through multiple stages.

Purchased





22.4 Equipment Status

Status	Description
Available	Ready for Booking
Reserved	Booking Approved
In Use	Currently Used
Under Maintenance	Technician Working
Calibration Due	Calibration Required
Out of Service	Equipment Failure
Retired	Permanently Removed

22.5 Equipment Categories

The platform should support unlimited categories.

Examples

- Mechanical
- Electrical
- Electronics
- Computer Science
- Biomedical
- Civil Engineering
- Chemical Engineering
- Physics
- Chemistry
- Biology

Each category may contain multiple equipment.

22.6 Equipment Master Information

Each equipment should contain the following information.

Basic Details

Equipment Name

Equipment Code

Serial Number

Manufacturer

Model Number

Category

Sub Category

Brand

Description

Purchase Date

Purchase Cost

Warranty Period

Current Value

Asset Number

QR Code

Barcode

Location Details

Institution

Department

Laboratory

Building

Floor

Room Number

Rack Number

GPS Location (Future)

Operational Details

Current Status

Available From

Maximum Booking Hours

Power Requirement

Operating Manual

Calibration Due Date

Maintenance Schedule

Assigned Technician

Ownership Details

Purchased By

Funding Source

Grant Number

Vendor

Invoice Number

AMC Provider

AMC Expiry

22.7 User Roles

Role	Access
Researcher	View Equipment
Faculty	View Equipment
Technician	Update Status
Lab Manager	Full CRUD
Institution Admin	Manage Inventory
System Admin	Global Inventory

22.8 Equipment Registration Workflow

Dashboard



Equipment Management



Add Equipment



Fill Details



Upload Documents



Generate QR Code



Save Equipment



Equipment Registered



Available for Booking

22.9 Equipment Search Workflow

Equipment List



Search



Filter



Sort



Equipment Details

22.10 Equipment Details Screen

Purpose

Displays complete equipment information.

Sections

General Information

Technical Information

Availability

Booking History

Maintenance History

Calibration

Documents

Analytics

Components

Equipment Image

Equipment Name

Status Badge

QR Code

Availability Calendar

Recent Bookings

Recent Maintenance

Download Manual

Book Button

22.11 Equipment List Screen

Purpose

Displays all available equipment.

Features

Search

Pagination

Filters

Sorting

Status Badges

Category Filter

Department Filter

Institution Filter

Export

Bulk Selection

22.12 Add Equipment Screen

Fields

Equipment Name

Equipment Code

Category

Manufacturer

Model Number

Purchase Date

Purchase Cost

Department

Laboratory

Technician

Warranty

Manual Upload

Image Upload

QR Code

Buttons

Save

Reset

Cancel

22.13 Edit Equipment Screen

Allows editing

Specifications

Location

Status

Documents

Images

Warranty

Technician

22.14 Equipment Availability Calendar

Displays

Daily Availability

Weekly Availability

Monthly Availability

Booked Slots

Maintenance Slots

Blocked Slots

Available Slots

22.15 Equipment Documents

Supported Files

PDF

DOCX

Images

Calibration Certificate

Warranty Card

Invoice

Operating Manual

Safety Guidelines

22.16 QR Code Workflow

Equipment Created

↓

Generate QR Code

↓

Store QR Image

↓

Print Sticker



Attach to Equipment



Scan



Equipment Details

22.17 Database Tables

equipment

Column	Type
id	BIGINT
equipment_code	VARCHAR
equipment_name	VARCHAR
category_id	BIGINT
laboratory_id	BIGINT
manufacturer	VARCHAR
model_number	VARCHAR
serial_number	VARCHAR
purchase_date	DATE
purchase_cost	DECIMAL
warranty_expiry	DATE
status	VARCHAR
qr_code	VARCHAR

image_url	VARCHAR
created_at	TIMESTAMP

equipment_category

Column	Type
id	BIGINT
category_name	VARCHAR
description	TEXT

equipment_documents

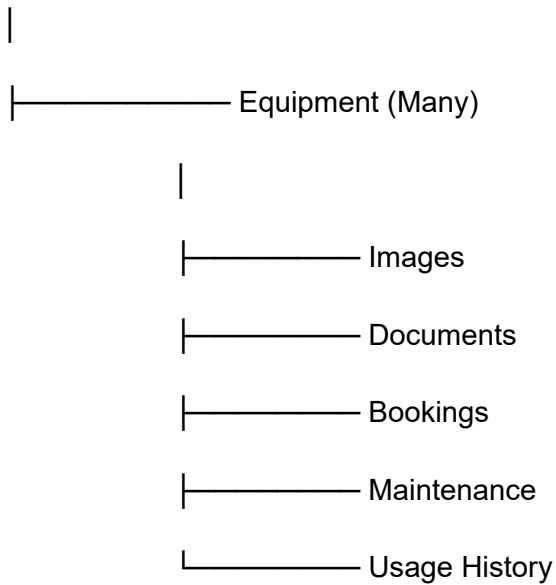
Column	Type
id	BIGINT
equipment_id	BIGINT
file_name	VARCHAR
document_type	VARCHAR
uploaded_at	TIMESTAMP

equipment_images

Column	Type
id	BIGINT
equipment_id	BIGINT
image_url	VARCHAR

22.18 Entity Relationship

Category (1)



22.19 REST APIs

Equipment APIs

Create Equipment

POST /api/equipment

Get Equipment

GET /api/equipment

Equipment Details

GET /api/equipment/{id}

Update Equipment

PUT /api/equipment/{id}

Delete Equipment

DELETE /api/equipment/{id}

Equipment Search

GET /api/equipment/search

Query Parameters

name

category

department

institution

status

Upload Equipment Image

POST /api/equipment/{id}/image

Upload Manual

POST /api/equipment/{id}/manual

Generate QR Code

POST /api/equipment/{id}/qr

22.20 Validation Rules

Equipment Name

Required

Maximum 200 characters

Equipment Code

Unique

Required

Purchase Cost

Must be positive

Purchase Date

Cannot be future date

Warranty Date

Must be greater than purchase date

Department

Required

Laboratory

Required

Image

Maximum 5MB

Manual

PDF Only

22.21 Business Rules

BR-021

Equipment Code must be unique.

BR-022

Equipment cannot exist without a laboratory.

BR-023

Deleted equipment cannot be restored.

BR-024

Only Available equipment can be booked.

BR-025

Equipment under maintenance cannot accept bookings.

BR-026

QR Code must be generated for every equipment.

BR-027

Equipment image is mandatory before publishing.

BR-028

Equipment documents must be version-controlled.

BR-029

Equipment category cannot be deleted if equipment exists.

BR-030

Retired equipment is read-only and cannot be modified.

22.22 Exception Handling

Exception	HTTP
Equipment Not Found	404
Duplicate Equipment Code	409
Invalid Category	400
Invalid Laboratory	400

Unauthorized	401
Forbidden	403
File Upload Failed	500

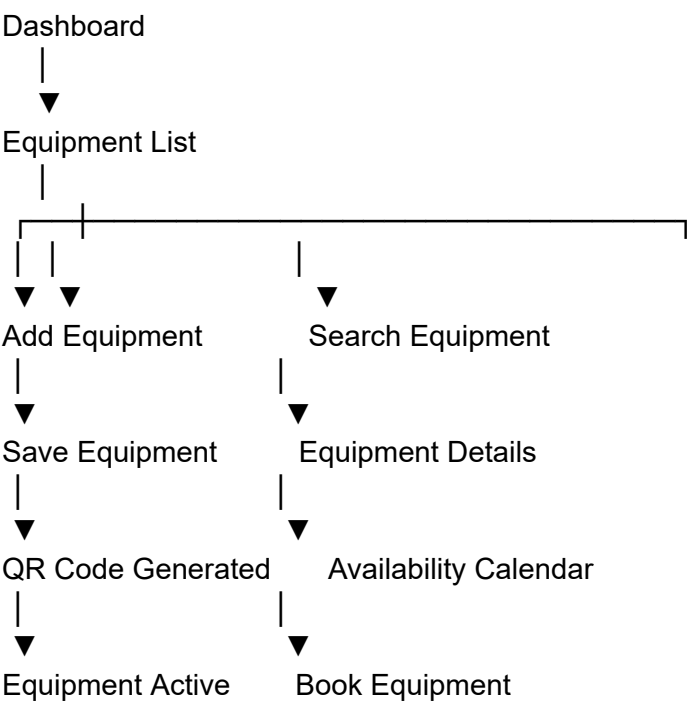
22.23 Acceptance Criteria

ID	Requirement
AC-031	Lab Manager can add equipment
AC-032	Equipment Code is unique
AC-033	QR Code is generated
AC-034	Equipment image uploads successfully
AC-035	Documents upload successfully
AC-036	Equipment search works with filters
AC-037	Status changes are reflected immediately
AC-038	Equipment details display correctly
AC-039	CRUD operations are audited
AC-040	Only authorized roles can modify inventory

22.24 Future Enhancements

- RFID-based asset tracking
 - IoT sensor integration for live status
 - AI-powered equipment recommendations
 - Automatic depreciation calculation
 - Predictive replacement planning
 - Digital twin visualization
 - Barcode scanner integration
 - Equipment health score
 - Spare parts inventory management
 - Vendor portal integration
-

22.25 Screen Flow



22.26 Screen Inventory

Screen ID	Screen Name
EQ-01	Equipment List
EQ-02	Equipment Details
EQ-03	Add Equipment
EQ-04	Edit Equipment
EQ-05	Equipment Calendar
EQ-06	Equipment Documents
EQ-07	Equipment Images
EQ-08	QR Code Preview
EQ-09	Equipment Analytics
EQ-10	Equipment History

✓ End of Part 5

At this point, your PRD has become a true enterprise document. We've fully specified the inventory module with workflows, data model, business rules, APIs, screens, and acceptance criteria.

Next (Part 6): Booking & Scheduling Module

This is the most complex functional module. We'll define:

- Booking workflow
- Calendar scheduling
- Approval/rejection process
- Waitlist management
- Booking conflict detection
- Recurring bookings
- Calendar UI
- Booking APIs
- Database schema
- State transitions
- Business rules
- Edge cases
- Notifications
- Sequence diagrams

This section will effectively describe how the platform manages equipment reservations from start to finish.

Excellent. We're now moving into the **intelligence layer** of the application. Unlike previous modules that focused on CRUD operations and workflows, this module provides **real-time insights and decision support** for administrators, lab managers, and institution leaders.

23. Module 4 – Booking & Scheduling Management

23.1 Module Overview

The Booking & Scheduling module enables users to reserve laboratory equipment for specific dates and time slots while ensuring that equipment conflicts, overlapping reservations, maintenance schedules, and institution policies are enforced.

The module provides a centralized booking workflow where users can search equipment, verify availability, submit booking requests, receive approvals, and track reservation history.

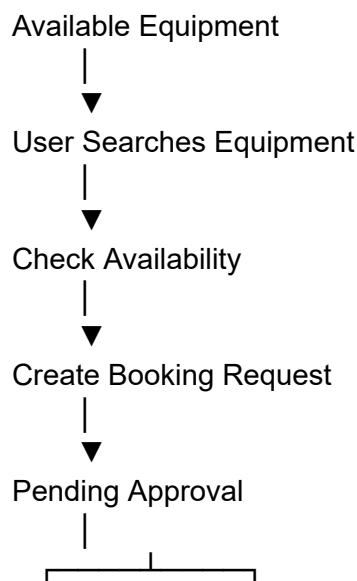
The scheduling engine ensures optimal utilization of laboratory equipment while preventing double-booking and resource conflicts.

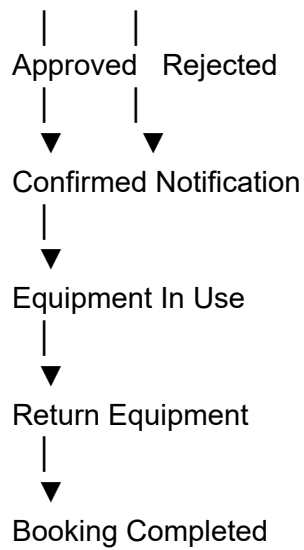
23.2 Module Objectives

The Booking Module shall:

- Allow equipment reservation.
 - Display equipment availability.
 - Prevent double booking.
 - Support approval workflow.
 - Support recurring bookings.
 - Manage waitlists.
 - Track booking lifecycle.
 - Send booking notifications.
 - Generate booking history.
 - Maintain booking audit logs.
-

23.3 Booking Lifecycle





23.4 Booking Status

Status	Description
Draft	Booking Created but not Submitted
Pending Approval	Awaiting Manager Approval
Approved	Booking Approved
Rejected	Booking Rejected
Confirmed	Reservation Confirmed
In Use	Equipment Currently Used
Completed	Equipment Returned
Cancelled	Booking Cancelled
Expired	Booking Time Passed
No Show	User Didn't Arrive

23.5 Booking Types

The platform supports multiple booking types.

Standard Booking

Single reservation.

Recurring Booking

Weekly laboratory sessions.

Example

Every Monday

10 AM – 12 PM

For 10 Weeks

Emergency Booking

Priority booking.

Requires Manager Approval.

Shared Booking

Equipment reserved by another institution.

23.6 Booking Workflow

Dashboard



Equipment Search



Equipment Details

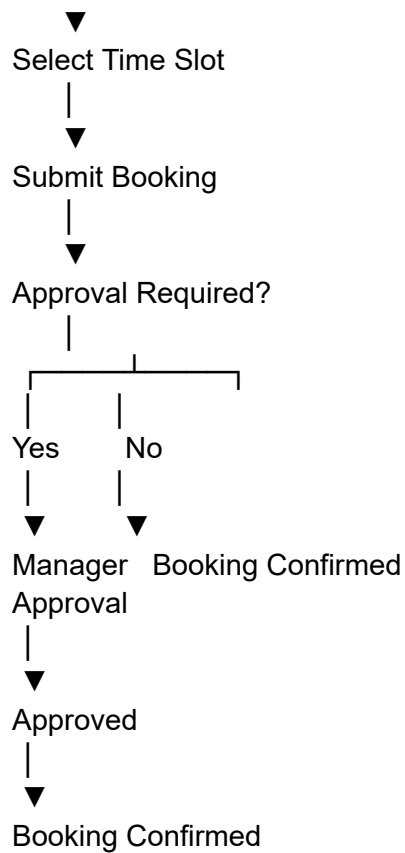


Availability Calendar



Select Date





23.7 Booking Screens

BK-01

Booking Calendar

Displays

- Daily View
- Weekly View
- Monthly View
- Equipment Availability

BK-02

Create Booking

Fields

Equipment

Booking Date

Start Time

End Time

Purpose

Research Project

Priority

Remarks

Buttons

Submit

Cancel

BK-03

Booking Details

Displays

Booking Information

Equipment Details

Approval Status

Booking Timeline

History

BK-04

Pending Approvals

Manager View

Shows

Pending Requests

Approve

Reject

Comments

BK-05

My Bookings

Researcher View

Displays

Upcoming

Completed

Cancelled

Rejected

BK-06

Booking History

Displays

Previous Reservations

Duration

Usage Hours

Status

23.8 Calendar View

The platform shall support multiple calendar views.

Daily

Weekly

Monthly

Timeline

Resource View

Equipment View

23.9 Booking Conflict Detection

The system must detect

- Time conflicts
- Equipment conflicts
- Maintenance conflicts
- Institution restrictions
- Department restrictions

Example

Existing Booking

10:00 - 12:00

↓

New Booking

11:00 - 1:00

↓

Conflict

↓

Reject Booking

23.10 Waitlist Management

If equipment is unavailable

↓

User joins waitlist

↓

Booking cancelled

↓

First waitlisted user notified

↓

Booking window opens

↓

Confirmation

23.11 Booking Approval Workflow

Researcher

↓

Booking Submitted

↓

Lab Manager

↓

Review Request

↓

Approve

↓

Booking Confirmed

↓

Notification Sent

Rejected

↓

Reason Added

↓

User Notified

23.12 Booking Rules

Examples

Maximum booking hours/day

Maximum active bookings/user

Advance booking limit

Cancellation deadline

Department priority

Institution priority

Emergency booking priority

23.13 Database Tables

bookings

Column	Type
id	BIGINT
equipment_id	BIGINT
user_id	BIGINT
booking_date	DATE

start_time	TIME
end_time	TIME
purpose	TEXT
booking_status	VARCHAR
approved_by	BIGINT
created_at	TIMESTAMP

booking_history

Column	Type
id	BIGINT
booking_id	BIGINT
status	VARCHAR
remarks	TEXT
updated_at	TIMESTAMP

booking_waitlist

Column	Type
id	BIGINT
equipment_id	BIGINT
user_id	BIGINT
position	INT

23.14 REST APIs

Create Booking

POST /api/bookings

Get Booking

GET /api/bookings/{id}

Get My Bookings

GET /api/bookings/my

Approve Booking

PUT /api/bookings/{id}/approve

Reject Booking

PUT /api/bookings/{id}/reject

Cancel Booking

PUT /api/bookings/{id}/cancel

Join Waitlist

POST /api/bookings/waitlist

23.15 Validation Rules

Equipment required

Date required

Start Time required

End Time required

Start Time < End Time

Equipment Available

User Active

Purpose Required

Booking Hours ≤ Maximum Allowed

23.16 Business Rules

BR-031

Equipment cannot be booked twice for the same time slot.

BR-032

Maintenance blocks booking automatically.

BR-033

Cancelled booking releases slot immediately.

BR-034

Recurring bookings require manager approval.

BR-035

Waitlist follows FIFO.

BR-036

Rejected bookings cannot be modified.

BR-037

Booking history cannot be deleted.

BR-038

Booking cancellation after equipment usage starts is not allowed.

BR-039

Only Managers can approve requests.

BR-040

Booking completion updates equipment availability automatically.

23.17 Notification Events

Notification triggers

Booking Created

Booking Approved

Booking Rejected

Booking Cancelled

Reminder (24 hours)

Reminder (1 hour)

Equipment Ready

Waitlist Promotion

Booking Completed

23.18 Exception Handling

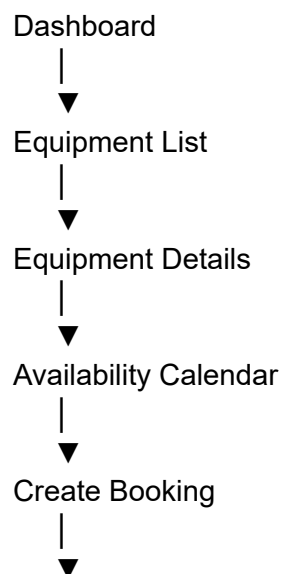
Exception	HTTP
Equipment Unavailable	409
Booking Conflict	409
Booking Not Found	404
Invalid Date	400
Unauthorized	401

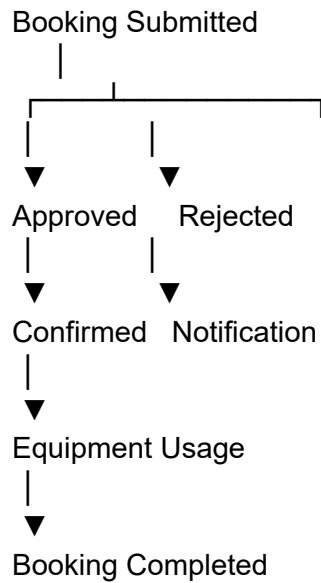
Forbidden	403
Server Error	500

23.19 Acceptance Criteria

ID	Requirement
AC-041	User can create booking
AC-042	Double booking prevented
AC-043	Calendar shows live availability
AC-044	Waitlist functions correctly
AC-045	Manager approves/rejects bookings
AC-046	Notifications sent on status changes
AC-047	Booking history maintained
AC-048	Equipment availability updates automatically
AC-049	Booking conflicts detected
AC-050	Booking reports generated successfully

23.20 Screen Flow





23.21 Screen Inventory

Screen ID	Screen Name
BK-01	Booking Calendar
BK-02	Create Booking
BK-03	Booking Details
BK-04	My Bookings
BK-05	Pending Approvals
BK-06	Booking History
BK-07	Waitlist
BK-08	Booking Analytics

23.22 Future Enhancements

- AI-based scheduling optimization
- Smart conflict resolution
- Automatic slot recommendations
- Google Calendar / Outlook integration
- QR-based equipment check-in and check-out

- Mobile booking support
 - Voice-based booking
 - Predictive demand forecasting
-

End of Part 6

The PRD now completely defines the booking engine, including workflows, scheduling rules, approval logic, APIs, validations, and UI.

Next (Part 7): Real-Time Utilization Monitoring & Analytics

This module will cover:

- Live equipment utilization tracking
- Utilization calculations
- Heatmaps
- KPI dashboards
- Idle equipment detection
- Department and institution analytics
- Charts and reports
- APIs and database design
- Business rules and acceptance criteria

This section will provide the intelligence layer of the platform, helping administrators make data-driven decisions.

24. Module 5 – Real-Time Utilization Monitoring & Analytics

24.1 Module Overview

The Real-Time Utilization Monitoring & Analytics module collects, processes, and visualizes operational data from equipment bookings, maintenance activities, and user interactions. It transforms raw data into actionable insights, enabling institutions to optimize equipment usage, reduce idle time, plan future investments, and improve operational efficiency.

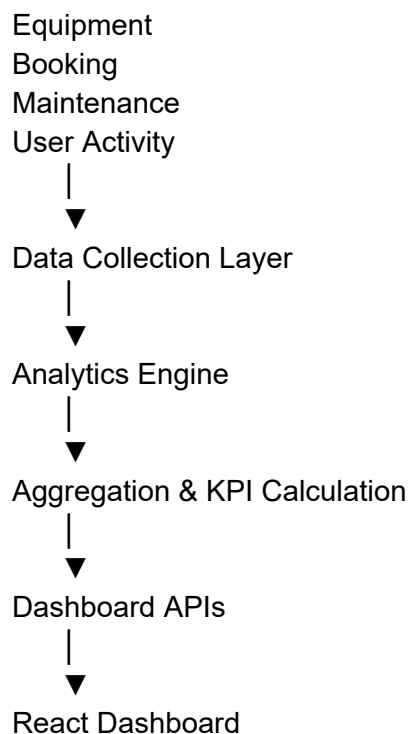
The module provides dashboards, KPIs, utilization trends, heatmaps, and predictive metrics for stakeholders at different organizational levels.

24.2 Module Objectives

The Analytics Module shall:

- Monitor real-time equipment utilization.
 - Calculate utilization percentages.
 - Detect idle equipment.
 - Analyze booking trends.
 - Compare departmental usage.
 - Generate KPI dashboards.
 - Provide downloadable reports.
 - Support decision-making for procurement.
 - Track maintenance impact on availability.
 - Offer predictive insights (future enhancement).
-

24.3 Analytics Architecture



24.4 Key Performance Indicators (KPIs)

KPI	Description
Equipment Utilization %	Percentage of booked time vs. available time
Idle Equipment	Equipment not used within a configurable period
Total Bookings	Number of completed and active bookings
Booking Success Rate	Approved bookings ÷ Total booking requests
Booking Rejection Rate	Rejected bookings ÷ Total booking requests
Maintenance Downtime	Total hours equipment was unavailable
Average Booking Duration	Mean reservation duration
Peak Usage Hours	Time periods with highest demand
Equipment Availability	Percentage of time equipment is available
Top Used Equipment	Most frequently reserved assets

24.5 User Roles

Role	Dashboard Access
Researcher	Personal Usage Dashboard
Faculty	Department Summary
Lab Manager	Laboratory Dashboard
Department Head	Department Analytics
Institution Admin	Institution Dashboard
System Admin	Global Dashboard

24.6 Dashboard Overview

The dashboard is divided into six major sections.

Dashboard Components

- KPI Cards
 - Utilization Charts
 - Equipment Status Summary
 - Booking Trends
 - Maintenance Statistics
 - Alerts & Notifications
-

24.7 KPI Cards

Each dashboard displays summary cards such as:

- Total Equipment
 - Active Equipment
 - Available Equipment
 - Equipment Under Maintenance
 - Active Bookings
 - Pending Approvals
 - Completed Bookings
 - Utilization Percentage
 - Idle Equipment Count
 - Upcoming Maintenance
-

24.8 Utilization Dashboard

Displays:

- Daily Utilization
- Weekly Utilization
- Monthly Utilization
- Quarterly Utilization
- Yearly Utilization

Metrics:

- Booking Hours
- Available Hours
- Utilization %

Formula:

Utilization % =
 $(\text{Total Booked Hours} \div \text{Total Available Hours}) \times 100$

Example:

Available Hours = 200

Booked Hours = 150

Utilization =

$150 / 200 \times 100$

= 75%

24.9 Equipment Heatmap

Purpose

Visualize periods of high and low equipment usage.

Legend

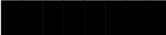
- Green → Low Usage
- Yellow → Medium Usage
- Orange → High Usage
- Red → Very High Usage

Example:

Monday

09:00 

10:00 

11:00 

12:00 

24.10 Booking Trend Analysis

Reports include:

- Daily bookings
- Weekly bookings
- Monthly bookings

- Semester trends
- Year-over-year comparison

Metrics

- New bookings
 - Completed bookings
 - Cancelled bookings
 - Rejected bookings
-

24.11 Equipment Performance Dashboard

Displays:

- Top 10 Most Used Equipment
 - Least Used Equipment
 - Frequently Cancelled Equipment
 - Equipment with Highest Downtime
 - Equipment Awaiting Maintenance
-

24.12 Department Analytics

Each department dashboard shows:

- Equipment Count
- Booking Count
- Utilization %
- Maintenance Hours
- Average Booking Duration
- Equipment Availability

Department comparison example:

Department	Utilization
Mechanical	88%
Electronics	74%

Computer Science	69%
Civil	53%

24.13 Institution Analytics

Displays:

- Institution-wise equipment distribution
 - Institution utilization
 - Active users
 - Total laboratories
 - Resource sharing statistics
 - Department rankings
-

24.14 Alerts & Notifications

Automatic alerts generated for:

- Equipment idle for more than 30 days
 - Utilization below threshold
 - Calibration due soon
 - Maintenance overdue
 - Booking rejection spike
 - Low equipment availability
 - Resource over-utilization
-

24.15 Screen Inventory

Screen ID	Screen Name
AN-01	Analytics Dashboard
AN-02	Utilization Dashboard
AN-03	Equipment Performance

AN-04	Department Analytics
AN-05	Institution Analytics
AN-06	Booking Trends
AN-07	KPI Dashboard
AN-08	Reports & Export

24.16 Database Views / Aggregated Tables

equipment_utilization_summary

Column	Type
equipment_id	BIGINT
booked_hours	DECIMAL
available_hours	DECIMAL
utilization_percentage	DECIMAL
last_updated	TIMESTAMP

booking_statistics

Column	Type
booking_date	DATE
total_bookings	INT
approved	INT
rejected	INT
completed	INT
cancelled	INT

department_utilization

Column	Type
department_id	BIGINT
utilization_percentage	DECIMAL
total_equipment	INT
booking_hours	DECIMAL

24.17 REST APIs

Dashboard Summary

GET /api/analytics/dashboard

Equipment Utilization

GET /api/analytics/utilization

Department Analytics

GET /api/analytics/departments

Institution Analytics

GET /api/analytics/institutions

Booking Trends

GET /api/analytics/bookings

Export Analytics Report

GET /api/analytics/export

Query Parameters:

format=pdf | excel | csv
dateRange=start,end
departmentId
institutionId

24.18 Validation Rules

- Date range must be valid.
 - End date cannot be earlier than start date.
 - Department must belong to selected institution.
 - Export format must be supported.
 - Users can only access analytics permitted by their role.
-

24.19 Business Rules

BR-041

Utilization is calculated only from completed and active bookings.

BR-042

Maintenance hours are excluded from available hours.

BR-043

Idle equipment is defined as no completed bookings for the configured threshold (default: 30 days).

BR-044

Department Heads can view only their department analytics.

BR-045

Institution Administrators cannot access analytics for other institutions.

BR-046

Reports are generated using the latest available aggregated data.

BR-047

Dashboard data refreshes automatically at configurable intervals (e.g., every 5 minutes).

BR-048

Analytics calculations must ignore deleted or retired equipment unless historical reporting is requested.

BR-049

Exports include only data the requesting user is authorized to access.

BR-050

Every report generation is logged for audit purposes.

24.20 Exception Handling

Exception	HTTP Status
Invalid Date Range	400
Unauthorized Access	401
Forbidden Analytics Scope	403
Department Not Found	404
Institution Not Found	404
Export Generation Failed	500

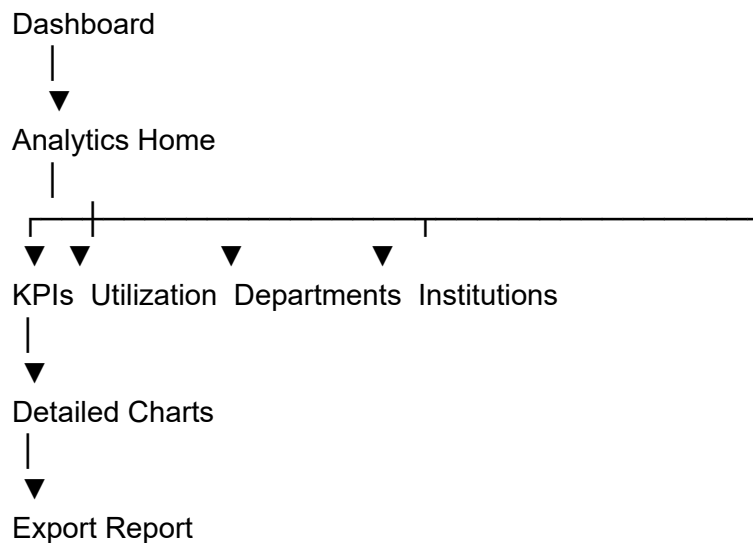
24.21 Acceptance Criteria

ID	Requirement
AC-051	Dashboard displays KPI cards correctly
AC-052	Utilization percentage is calculated accurately
AC-053	Heatmap reflects booking activity
AC-054	Department analytics display correct metrics
AC-055	Institution analytics aggregate departmental data
AC-056	Reports export successfully in supported formats
AC-057	Idle equipment alerts trigger correctly
AC-058	Dashboard respects role-based access
AC-059	Auto-refresh updates dashboard data
AC-060	Audit logs record report generation

24.22 Future Enhancements

- AI-powered demand forecasting
 - Predictive maintenance recommendations
 - Equipment utilization forecasting
 - Cost optimization suggestions
 - Carbon footprint analysis
 - Digital twin dashboards
 - Machine learning anomaly detection
 - Interactive drill-down dashboards
 - Real-time IoT sensor integration
 - Executive mobile analytics dashboard
-

24.23 Screen Flow



✓ End of Part 7

The PRD now includes a complete analytics module with KPI definitions, dashboards, reporting, data aggregation, APIs, business rules, and acceptance criteria.

Next (Part 8): Maintenance & Calibration Management

This module will document:

- Preventive and corrective maintenance workflows
- Calibration scheduling
- Work order management

- Technician assignment
- Spare parts tracking
- Maintenance history
- Downtime calculations
- APIs, database schema, UI screens, validations, business rules, and acceptance criteria

This module ensures equipment reliability and regulatory compliance throughout its operational lifecycle.

Excellent. This module distinguishes the platform from a standard Laboratory Management System (LMS). It enables **multiple universities, research institutions, and laboratories** to securely share high-value equipment, maximizing utilization while reducing procurement costs.

25. Module 6 – Maintenance & Calibration Management

25.1 Module Overview

The Maintenance & Calibration Management module manages the complete lifecycle of equipment servicing, preventive maintenance, corrective repairs, calibration schedules, spare parts usage, and technician assignments. It minimizes equipment downtime, ensures compliance with laboratory standards, and improves equipment reliability.

This module integrates closely with the Equipment Inventory, Booking, and Analytics modules to automatically block unavailable equipment and track maintenance-related KPIs.

25.2 Module Objectives

The Maintenance Module shall:

- Schedule preventive maintenance.
- Handle corrective maintenance requests.
- Manage calibration schedules.
- Assign technicians to work orders.

- Track maintenance history.
 - Monitor equipment downtime.
 - Record spare parts usage.
 - Block equipment during maintenance.
 - Notify stakeholders about upcoming maintenance.
 - Generate maintenance reports.
-

25.3 Maintenance Types

Preventive Maintenance

Scheduled maintenance performed at regular intervals to reduce the likelihood of equipment failure.

Examples:

- Monthly inspection
 - Quarterly servicing
 - Annual overhaul
-

Corrective Maintenance

Maintenance performed after an equipment failure.

Examples:

- Motor replacement
 - Sensor repair
 - Electrical fault correction
-

Calibration

Periodic verification of measurement accuracy using certified standards.

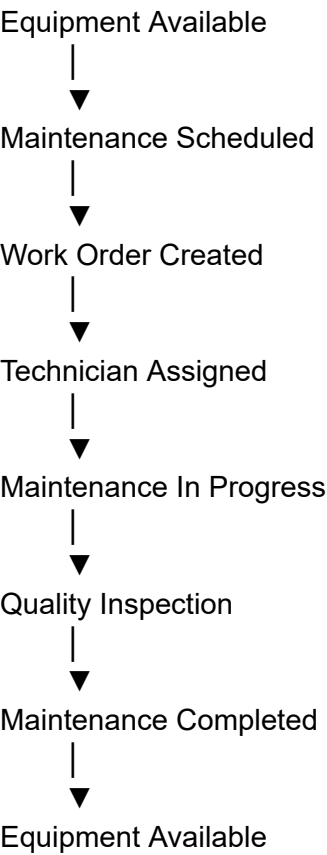
Examples:

- Oscilloscope calibration
 - Digital weighing scale calibration
 - Pressure gauge calibration
-

Emergency Maintenance

Immediate maintenance performed due to critical equipment failure affecting laboratory operations.

25.4 Maintenance Lifecycle



25.5 Work Order Lifecycle

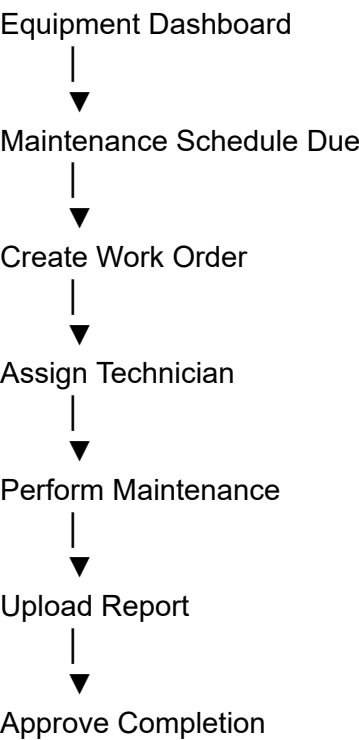
Status	Description
Created	Work order generated
Assigned	Technician assigned
Accepted	Technician accepted task
In Progress	Maintenance ongoing

Waiting for Parts	Spare parts required
Under Inspection	QA verification
Completed	Successfully finished
Cancelled	Work order cancelled

25.6 User Roles

Role	Responsibilities
Technician	Perform maintenance, update work orders
Lab Manager	Create work orders, assign technicians
Institution Admin	Monitor maintenance activities
System Admin	Configure maintenance policies
Researcher	View maintenance status only

25.7 Preventive Maintenance Workflow



↓
Equipment Available

25.8 Corrective Maintenance Workflow

Equipment Failure Reported
↓
Create Incident
↓
Generate Work Order
↓
Assign Technician
↓
Diagnosis
↓
Repair Equipment
↓
Testing
↓
Close Work Order

25.9 Calibration Workflow

Calibration Due
↓
Schedule Calibration
↓
Assign Technician / External Vendor
↓
Perform Calibration
↓

Upload Calibration Certificate



Update Next Calibration Date

25.10 Screens

MT-01 – Maintenance Dashboard

Displays:

- Upcoming maintenance
 - Overdue maintenance
 - Active work orders
 - Equipment under maintenance
 - Calibration due
 - Maintenance KPIs
-

MT-02 – Work Order List

Features:

- Search
 - Filter
 - Technician assignment
 - Status tracking
 - Priority filter
-

MT-03 – Create Work Order

Fields:

- Equipment
- Maintenance Type
- Priority
- Description
- Scheduled Date
- Technician
- Estimated Duration

Buttons:

- Save
 - Assign
 - Cancel
-

MT-04 – Work Order Details

Displays:

- Equipment details
 - Assigned technician
 - Timeline
 - Maintenance history
 - Spare parts used
 - Attachments
-

MT-05 – Calibration Dashboard

Displays:

- Upcoming calibrations
 - Expired calibrations
 - Certificates
 - Next due dates
-

MT-06 – Maintenance History

Displays:

- Previous work orders
 - Repair cost
 - Downtime
 - Technician
 - Reports
-

25.11 Spare Parts Management

Each maintenance activity may consume spare parts.

Track:

- Part Number
 - Part Name
 - Quantity Used
 - Unit Cost
 - Vendor
 - Remaining Stock
-

25.12 Notifications

Automatic notifications for:

- Maintenance scheduled
 - Work order assigned
 - Maintenance overdue
 - Calibration due in 7 days
 - Calibration overdue
 - Work order completed
 - Equipment available again
-

25.13 Database Tables

maintenance_work_orders

Column	Type
id	BIGINT
equipment_id	BIGINT
maintenance_type	VARCHAR
priority	VARCHAR
assigned_to	BIGINT
status	VARCHAR

scheduled_date	DATE
completion_date	DATE
remarks	TEXT

maintenance_history

Column	Type
id	BIGINT
equipment_id	BIGINT
work_order_id	BIGINT
maintenance_type	VARCHAR
downtime_hours	DECIMAL
total_cost	DECIMAL
completed_by	BIGINT

calibration_records

Column	Type
id	BIGINT
equipment_id	BIGINT
calibration_date	DATE
next_due_date	DATE
certificate_url	VARCHAR
calibrated_by	VARCHAR

spare_parts_usage

Column	Type
id	BIGINT

work_order_id	BIGINT
part_number	VARCHAR
quantity_used	INT
unit_cost	DECIMAL

25.14 REST APIs

Create Work Order

POST /api/maintenance/work-orders

Get Work Orders

GET /api/maintenance/work-orders

Update Work Order

PUT /api/maintenance/work-orders/{id}

Complete Work Order

PUT /api/maintenance/work-orders/{id}/complete

Schedule Calibration

POST /api/calibration

Upload Calibration Certificate

POST /api/calibration/{id}/certificate

Maintenance History

GET /api/maintenance/history

25.15 Validation Rules

- Equipment must exist.
 - Maintenance type is mandatory.
 - Scheduled date cannot be in the past (unless creating historical records).
 - Completion date cannot precede scheduled date.
 - Technician must be active.
 - Calibration certificate upload supports PDF only.
 - Spare part quantity must be greater than zero.
 - Estimated duration must be positive.
-

25.16 Business Rules

BR-051

Equipment under maintenance cannot be booked.

BR-052

Only assigned technicians can update work orders.

BR-053

A work order cannot be closed until all mandatory fields are completed.

BR-054

Calibration automatically updates the next due date based on equipment policy.

BR-055

Preventive maintenance schedules are generated automatically according to configured intervals.

BR-056

Maintenance completion automatically changes equipment status to **Available**, unless inspection fails.

BR-057

Every maintenance action is recorded in the audit log.

BR-058

Work orders cannot be deleted after completion.

BR-059

Maintenance cost contributes to lifecycle cost analytics.

BR-060

Calibration certificates must remain accessible for audit purposes.

25.17 Exception Handling

Exception	HTTP Status
Equipment Not Found	404
Work Order Not Found	404
Invalid Technician	400
Equipment Already Under Maintenance	409
Unauthorized	401
Forbidden	403
File Upload Failed	500

25.18 Acceptance Criteria

ID	Requirement
AC-061	Preventive maintenance can be scheduled
AC-062	Corrective maintenance work orders can be created
AC-063	Calibration schedules are tracked
AC-064	Equipment is blocked during maintenance
AC-065	Work order status updates correctly
AC-066	Maintenance history is retained
AC-067	Calibration certificates can be uploaded
AC-068	Notifications are sent for due and overdue maintenance
AC-069	Downtime metrics are reflected in analytics
AC-070	Audit logs capture all maintenance events

25.19 Maintenance KPI Dashboard

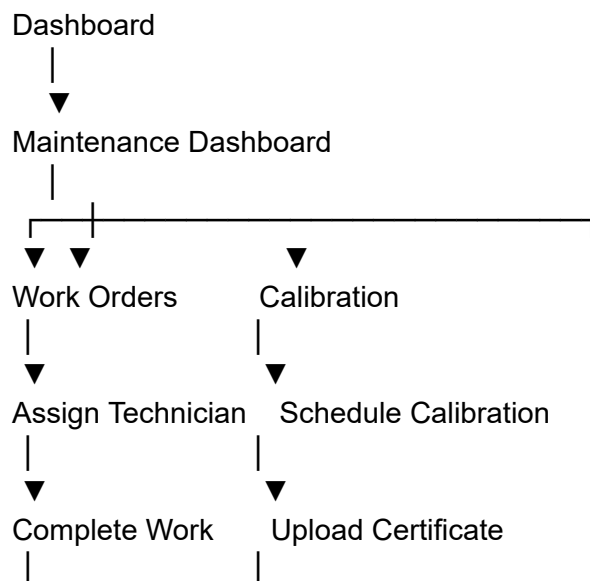
The Maintenance Dashboard displays:

- Total Active Work Orders
 - Preventive Maintenance Compliance (%)
 - Corrective Maintenance Count
 - Average Repair Time (MTTR)
 - Mean Time Between Failures (MTBF)
 - Total Downtime (Hours)
 - Maintenance Cost by Equipment
 - Calibration Compliance (%)
 - Overdue Maintenance Count
 - Technician Workload Distribution
-

25.20 Future Enhancements

- IoT-based predictive maintenance
 - AI-powered failure prediction
 - QR code scanning for instant work order creation
 - Mobile technician application
 - Digital maintenance checklists
 - Vendor service portal integration
 - Automatic spare parts inventory updates
 - Electronic signature for maintenance approval
 - Integration with ERP/CMMS systems
 - Predictive spare parts demand forecasting
-

25.21 Screen Flow



End of Part 8

Your PRD now covers:

- Authentication & Authorization
- Institution Management
- Equipment Inventory
- Booking & Scheduling
- Real-Time Analytics
- **Maintenance & Calibration Management**

Next (Part 9): Inter-Institution Resource Sharing

This module will define how multiple universities and research institutions securely share laboratory equipment, including:

- Cross-institution booking workflows
- Approval chains
- Resource-sharing agreements
- Cost allocation and billing
- Access control
- Availability synchronization
- APIs, database schema, business rules, and UI specifications.

26. Module 7 – Inter-Institution Resource Sharing

26.1 Module Overview

The Inter-Institution Resource Sharing module enables registered institutions to share laboratory equipment with partner institutions through a secure and policy-driven workflow. It supports cross-institution equipment discovery, booking requests, approval workflows, cost allocation, usage tracking, and access control.

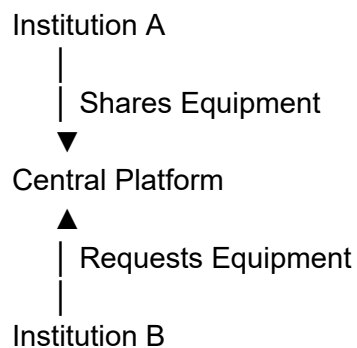
The module promotes collaboration by allowing institutions to use specialized equipment that they may not own, thereby improving utilization and reducing duplicate investments.

26.2 Module Objectives

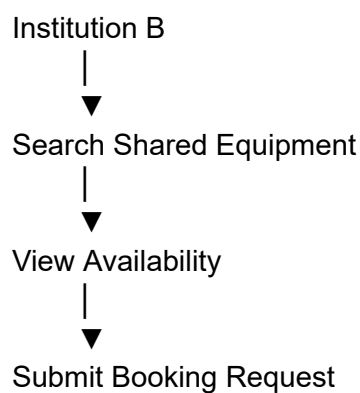
The Resource Sharing Module shall:

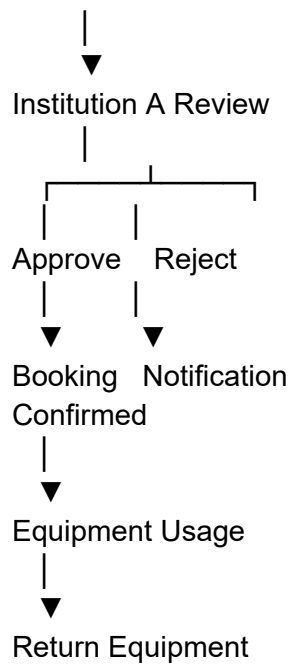
- Enable cross-institution equipment sharing.
 - Support institution partnership management.
 - Allow external booking requests.
 - Enforce sharing policies.
 - Manage approval workflows.
 - Allocate usage costs.
 - Track shared equipment usage.
 - Synchronize availability across institutions.
 - Generate collaboration reports.
 - Maintain complete audit trails.
-

26.3 Resource Sharing Model



26.4 Sharing Workflow





26.5 Sharing Policies

Each institution can configure:

- Public Sharing
 - Partner-Only Sharing
 - Department-Restricted Sharing
 - Approval Required
 - Hourly Usage Charges
 - Daily Usage Charges
 - Security Deposit
 - Maximum Booking Duration
 - Booking Notice Period
 - Cancellation Policy
-

26.6 User Roles

Role	Responsibilities
External Researcher	Search and request shared equipment
Institution Admin	Configure sharing policies

Lab Manager	Approve or reject external requests
Finance Officer	Review usage charges
System Admin	Manage institution partnerships

26.7 Institution Partnership Management

Administrators can:

- Create partnership agreements.
 - Invite partner institutions.
 - Accept or reject partnership requests.
 - Define agreement validity.
 - Configure equipment categories available for sharing.
 - Suspend or terminate partnerships.
-

26.8 Shared Equipment Catalog

Each shared equipment entry includes:

- Equipment Name
 - Equipment Code
 - Category
 - Institution
 - Laboratory
 - Availability Status
 - Booking Calendar
 - Usage Cost
 - Security Deposit
 - Operating Restrictions
 - Required Training
 - Contact Person
-

26.9 Screens

RS-01 – Shared Equipment Search

Features:

- Search
 - Advanced Filters
 - Institution Filter
 - Category Filter
 - Availability Filter
 - Cost Filter
-

RS-02 – Equipment Details

Displays:

- Specifications
 - Availability Calendar
 - Institution Information
 - Usage Charges
 - Booking Button
 - Documents
 - Safety Instructions
-

RS-03 – External Booking Request

Fields:

- Equipment
- Institution
- Research Purpose
- Booking Date
- Start Time
- End Time
- Principal Investigator
- Funding Source
- Remarks

Buttons:

- Submit
 - Cancel
-

RS-04 – Approval Dashboard

Displays:

- Pending Requests
 - Institution Name
 - Requested Duration
 - Cost Estimate
 - Approve
 - Reject
 - Add Comments
-

RS-05 – Partnership Management

Displays:

- Active Partners
 - Pending Invitations
 - Agreement Expiry
 - Shared Categories
 - Status
-

26.10 Cost Allocation

The system supports:

- Free sharing
- Hourly billing
- Daily billing
- Fixed booking charges
- Security deposits
- Damage recovery costs

Example:

Usage Time = 5 Hours

Hourly Rate = ₹500

Total Charge

= 5 × 500

= ₹2500

26.11 Availability Synchronization

Whenever equipment is:

- Booked
- Under maintenance
- Reserved
- Retired

its availability is synchronized across all partner institutions to prevent conflicting reservations.

26.12 Database Tables

institution_partnerships

Column	Type
id	BIGINT
institution_a_id	BIGINT
institution_b_id	BIGINT
agreement_start	DATE
agreement_end	DATE
status	VARCHAR

shared_equipment

Column	Type
id	BIGINT
equipment_id	BIGINT
hourly_rate	DECIMAL
daily_rate	DECIMAL
security_deposit	DECIMAL

sharing_status VARCHAR

external_booking_requests

Column	Type
id	BIGINT
shared_equipment_id	BIGINT
requesting_institution_id	BIGINT
requested_by	BIGINT
booking_date	DATE
start_time	TIME
end_time	TIME
status	VARCHAR

sharing_transactions

Column	Type
id	BIGINT
booking_request_id	BIGINT
total_amount	DECIMAL
payment_status	VARCHAR
created_at	TIMESTAMP

26.13 REST APIs

Search Shared Equipment

GET /api/shared-equipment

Equipment Details

GET /api/shared-equipment/{id}

Create External Booking

POST /api/shared-bookings

Approve External Booking

PUT /api/shared-bookings/{id}/approve

Reject External Booking

PUT /api/shared-bookings/{id}/reject

Partnership Management

POST /api/partnerships

GET /api/partnerships

PUT /api/partnerships/{id}

DELETE /api/partnerships/{id}

Cost Calculation

GET /api/shared-bookings/{id}/cost

26.14 Validation Rules

- Institutions must have an active partnership.
 - Agreement must not be expired.
 - Equipment must be marked as shareable.
 - Equipment must be available.
 - Booking dates must not overlap.
 - Security deposit must be non-negative.
 - Hourly and daily rates must be valid.
-

26.15 Business Rules

BR-061

Only institutions with an active partnership can access shared equipment.

BR-062

Only equipment marked as "Shareable" appears in the shared catalog.

BR-063

External bookings always require approval from the owning institution unless auto-approval is configured.

BR-064

Equipment under maintenance cannot be shared.

BR-065

Sharing costs are calculated based on the institution's configured pricing policy.

BR-066

Agreement expiration immediately disables new sharing requests.

BR-067

Availability changes are synchronized in real time across partner institutions.

BR-068

Damage reports must be recorded before closing external bookings if applicable.

BR-069

Every external transaction generates an audit log.

BR-070

Institution Administrators can suspend sharing at any time.

26.16 Exception Handling

Exception	HTTP Status
Partnership Not Found	404
Partnership Expired	403
Equipment Not Shareable	400
External Booking Conflict	409
Unauthorized	401
Forbidden	403

26.17 Acceptance Criteria

ID	Requirement
AC-071	Partner institutions can search shared equipment
AC-072	External booking requests can be submitted
AC-073	Approval workflow functions correctly
AC-074	Availability synchronizes across institutions
AC-075	Sharing charges are calculated accurately
AC-076	Partnership rules are enforced
AC-077	External booking history is maintained
AC-078	Notifications are sent on approval/rejection
AC-079	Audit logs record all sharing activities
AC-080	Reports include inter-institution usage statistics

26.18 Analytics

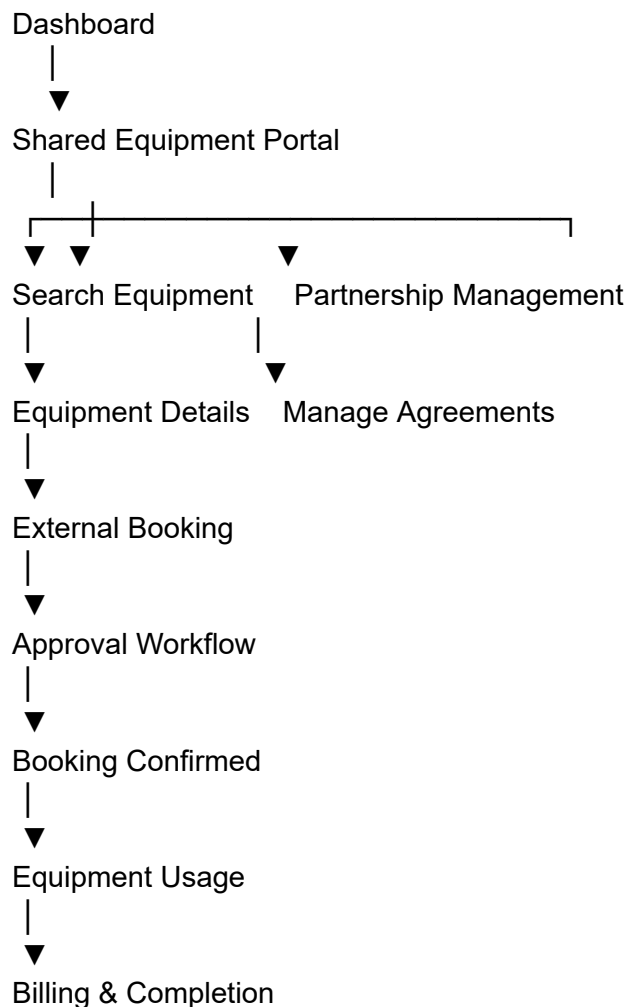
Resource Sharing Dashboard displays:

- Total Partner Institutions
 - Active Sharing Agreements
 - Shared Equipment Count
 - External Bookings
 - Revenue Generated
 - Cost Savings
 - Top Shared Equipment
 - Institution Collaboration Index
 - Average Approval Time
 - External Utilization Percentage
-

26.19 Future Enhancements

- National research equipment exchange network
 - AI-based institution matching
 - Blockchain-backed sharing agreements
 - Smart contract-based billing
 - Cross-border collaboration support
 - Automated insurance verification
 - Federated identity (SAML/OIDC)
 - Shared equipment recommendation engine
 - Carbon savings dashboard through shared utilization
 - Government-funded equipment sharing integration
-

26.20 Screen Flow



✓ End of Part 9

The PRD now defines a complete **multi-institution collaboration framework**, including partnership management, external bookings, cost allocation, synchronization, and governance.

Next (Part 10): Notification & Communication Management

This module will cover:

- Email, SMS, and in-app notifications
- Notification templates
- Event-driven notification engine
- Reminder scheduling
- Escalation rules
- User notification preferences
- Messaging APIs
- Delivery tracking
- Business rules and acceptance criteria

This module will ensure that users stay informed at every stage of the platform's workflows.

27. Module 8 – Notification & Communication Management

27.1 Module Overview

The Notification & Communication Management module provides a centralized event-driven communication system for the Lab Resource Utilization Platform. It delivers real-time notifications, reminders, alerts, and announcements through multiple channels, including in-app notifications, email, SMS, and push notifications (future enhancement).

The module improves user engagement, reduces missed bookings, ensures timely maintenance, and supports operational transparency.

27.2 Module Objectives

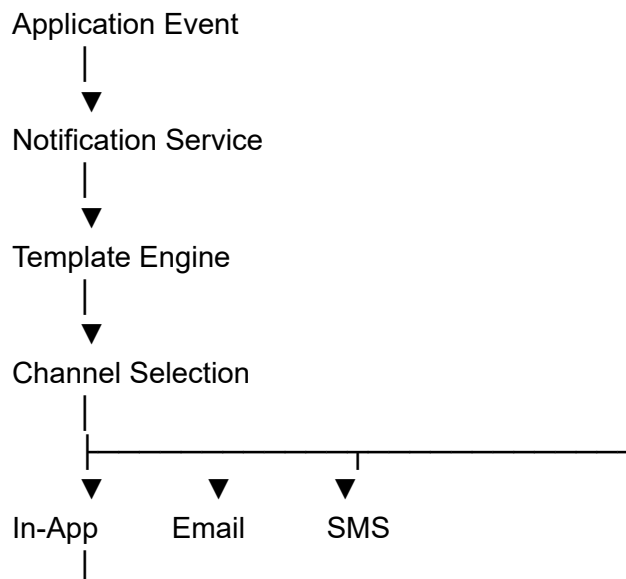
The Notification Module shall:

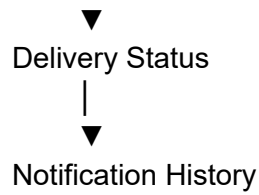
- Send real-time notifications.
 - Deliver booking reminders.
 - Notify users of approvals and rejections.
 - Alert technicians about maintenance tasks.
 - Send calibration reminders.
 - Broadcast institution-wide announcements.
 - Track notification delivery.
 - Allow users to manage notification preferences.
 - Retry failed notifications.
 - Maintain notification history.
-

27.3 Notification Channels

Channel	Description
In-App Notification	Displayed within the application
Email	Sent to registered email address
SMS	Sent to registered mobile number
Push Notification (<i>Future</i>)	Mobile application notifications
Webhook (<i>Future</i>)	Integration with external systems

27.4 Event-Driven Notification Flow





27.5 Notification Events

Authentication

- User Registration Successful
 - Email Verification
 - Password Reset
 - Login from New Device
 - Account Locked
 - Password Changed
-

Equipment

- New Equipment Added
 - Equipment Updated
 - Equipment Retired
 - Equipment Status Changed
-

Booking

- Booking Created
 - Booking Approved
 - Booking Rejected
 - Booking Cancelled
 - Booking Reminder (24 Hours)
 - Booking Reminder (1 Hour)
 - Equipment Ready
 - Booking Completed
 - Waitlist Promoted
-

Maintenance

- Work Order Created
- Technician Assigned

- Maintenance Started
 - Maintenance Completed
 - Calibration Due
 - Calibration Overdue
-

Institution

- Partnership Invitation
 - Partnership Approved
 - New Announcement
 - Policy Updated
-

27.6 Notification Templates

Each notification template contains:

- Subject
- Title
- Message
- Priority
- Channel
- Language
- Variables
- Status

Example:

Booking Approved

Subject:

Booking Approved

Message:

Your booking for {{equipmentName}} has been approved.

Booking Date:
{{bookingDate}}

Time:
{{startTime}} - {{endTime}}

27.7 User Notification Preferences

Users can configure:

- Enable Email
 - Enable SMS
 - Enable In-App
 - Enable Push Notifications
 - Reminder Frequency
 - Quiet Hours
 - Language Preference
-

27.8 Notification Priority

Priority	Description
Critical	Immediate delivery
High	Delivered instantly
Medium	Delivered within a few minutes
Low	Batched delivery

27.9 Screens

NT-01 – Notification Center

Displays:

- All notifications
 - Read / Unread status
 - Filters
 - Search
 - Delete
 - Mark as Read
-

NT-02 – Notification Preferences

Allows users to configure:

- Email Notifications
 - SMS Notifications
 - In-App Notifications
 - Reminder Settings
 - Language
-

NT-03 – Announcement Center

Displays:

- Institution Announcements
 - Department Announcements
 - System Announcements
-

NT-04 – Notification History

Displays:

- Sent Time
 - Delivery Status
 - Channel
 - Retry Count
 - Open Status
-

27.10 Notification Workflow

Booking Approved



Generate Event



Load Template



Replace Variables



Check User Preferences



Send Notification



Update Delivery Status

27.11 Reminder Engine

Automatic reminders for:

- Upcoming bookings (24 hours before)
 - Upcoming bookings (1 hour before)
 - Maintenance due
 - Calibration due
 - Partnership expiry
 - Warranty expiry
 - Annual compliance checks
-

27.12 Announcement Management

Institution Administrators can create announcements with:

- Title
- Description
- Audience
- Start Date
- End Date
- Priority
- Attachments

Target audiences:

- All Users
 - Institution
 - Department
 - Laboratory
 - Specific Roles
-

27.13 Database Tables

notifications

Column	Type
id	BIGINT
user_id	BIGINT
title	VARCHAR
message	TEXT
notification_type	VARCHAR
priority	VARCHAR
status	VARCHAR
created_at	TIMESTAMP

notification_templates

Column	Type
id	BIGINT
template_name	VARCHAR
subject	VARCHAR
body	TEXT
channel	VARCHAR
language	VARCHAR

notification_preferences

Column	Type
id	BIGINT
user_id	BIGINT

email_enabled	BOOLEAN
sms_enabled	BOOLEAN
in_app_enabled	BOOLEAN
language	VARCHAR

announcements

Column	Type
id	BIGINT
title	VARCHAR
description	TEXT
audience	VARCHAR
start_date	DATE
end_date	DATE
created_by	BIGINT

27.14 REST APIs

Get Notifications

GET /api/notifications

Mark Notification as Read

PUT /api/notifications/{id}/read

Delete Notification

DELETE /api/notifications/{id}

Update Preferences

PUT /api/notifications/preferences

Create Announcement

POST /api/announcements

Get Announcements

GET /api/announcements

27.15 Validation Rules

- Notification title is mandatory.
 - Message body cannot be empty.
 - Announcement end date must be after start date.
 - User preferences must exist before sending notifications.
 - Only authorized users can create announcements.
 - Templates must support required variables.
-

27.16 Business Rules

BR-071

Notifications must respect user communication preferences.

BR-072

Critical notifications cannot be disabled by users.

BR-073

Duplicate notifications for the same event should be prevented.

BR-074

Failed deliveries must be retried based on configurable retry policies.

BR-075

Expired announcements are automatically archived.

BR-076

Unread notifications remain available until deleted or archived.

BR-077

Notification templates support localization for multiple languages.

BR-078

Delivery status is recorded for every notification channel.

BR-079

Only Institution Administrators and System Administrators can publish announcements.

BR-080

Notification history must be retained for audit and troubleshooting.

27.17 Exception Handling

Exception	HTTP Status
Notification Not Found	404
Invalid Template	400
Invalid User Preference	400
Delivery Failure	500
Unauthorized	401
Forbidden	403

27.18 Acceptance Criteria

ID	Requirement
AC-081	Notifications are generated for supported events
AC-082	User preferences are respected
AC-083	Booking reminders are delivered on schedule
AC-084	Maintenance alerts reach assigned technicians
AC-085	Announcements are visible to the intended audience
AC-086	Notification history is available
AC-087	Failed notifications are retried

AC-088 Delivery status is tracked

AC-089 Critical alerts bypass optional notification settings

AC-090 Audit logs capture notification events

27.19 Notification Analytics

The Notification Dashboard displays:

- Notifications Sent
 - Delivery Success Rate
 - Email Delivery Rate
 - SMS Delivery Rate
 - In-App Read Rate
 - Failed Deliveries
 - Average Delivery Time
 - Most Common Notification Types
 - User Engagement Rate
 - Announcement Reach
-

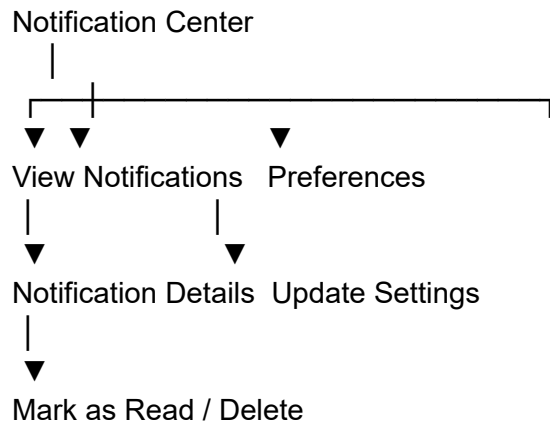
27.20 Future Enhancements

- Mobile push notifications
 - Microsoft Teams integration
 - Slack integration
 - WhatsApp Business notifications
 - AI-powered notification scheduling
 - Rich media notifications (images/videos)
 - Voice call alerts for critical events
 - Read receipt analytics
 - Smart notification grouping
 - Notification fatigue detection
-

27.21 Screen Flow

Dashboard





✓ End of Part 10

The PRD now includes a comprehensive **Notification & Communication Management** module with event-driven architecture, templates, user preferences, reminders, announcements, APIs, business rules, and analytics.

Next (Part 11): Reporting, Billing & Audit Management

The next module will consolidate operational reporting, billing for shared resources, financial tracking, and audit logging. It will cover:

- Report generation and exports (PDF, Excel, CSV)
- Billing and invoicing for inter-institution sharing
- Payment tracking
- Audit logs and compliance
- Financial dashboards
- Scheduled reports
- Database schema
- APIs
- Business rules
- Acceptance criteria

This module completes the governance and compliance capabilities of the platform.

28. Module 9 – Reporting, Billing & Audit Management

28.1 Module Overview

The Reporting, Billing & Audit Management module provides comprehensive operational reporting, financial management for shared resource usage, and immutable audit logging for all significant system activities.

It enables administrators to generate reports, calculate usage-based charges, issue invoices, track payments, monitor compliance, and maintain detailed audit trails for security and governance.

This module integrates with:

- Booking Management
 - Resource Sharing
 - Maintenance
 - Analytics
 - Notification
 - Authentication
-

28.2 Module Objectives

The module shall:

- Generate operational reports.
 - Generate financial reports.
 - Calculate equipment usage charges.
 - Generate invoices.
 - Track payments.
 - Export reports.
 - Maintain audit logs.
 - Support compliance audits.
 - Schedule recurring reports.
 - Provide financial dashboards.
-

28.3 Reporting Categories

Operational Reports

- Equipment Inventory Report
- Equipment Utilization Report
- Booking Report
- Booking Cancellation Report
- Booking Rejection Report

- Maintenance Report
 - Calibration Report
 - Technician Performance Report
-

Administrative Reports

- Institution Summary
 - Department Summary
 - Laboratory Summary
 - User Activity Report
 - Login Activity Report
 - Role Assignment Report
-

Financial Reports

- Equipment Usage Charges
 - Revenue Report
 - Outstanding Payments
 - Invoice Report
 - Institution Billing Summary
 - Resource Sharing Revenue
-

Compliance Reports

- Audit Trail
 - Data Access Report
 - Security Events
 - Configuration Changes
 - User Permission Changes
-

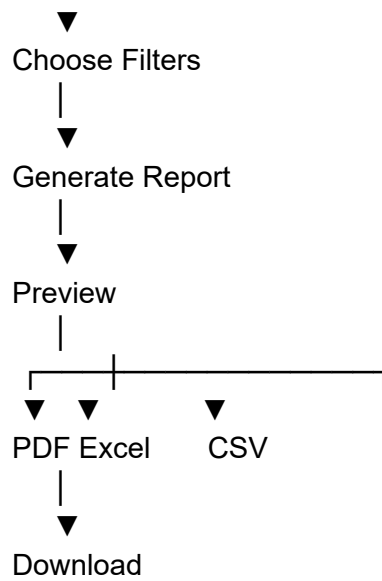
28.4 Reporting Workflow

Dashboard

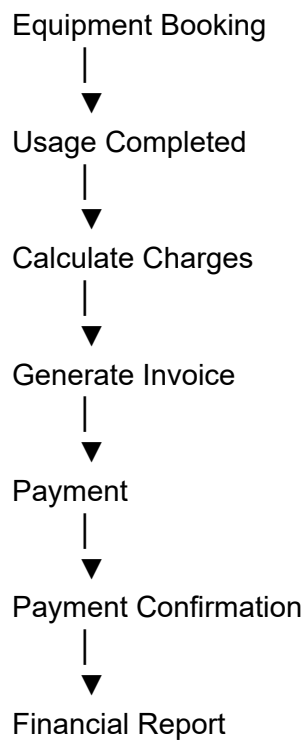


Select Report





28.5 Billing Workflow



28.6 Pricing Models

The platform supports:

- Hourly Pricing

- Daily Pricing
 - Weekly Pricing
 - Monthly Pricing
 - Flat Rate
 - Security Deposit
 - Penalty Charges
 - Damage Charges
-

28.7 Invoice Components

Each invoice contains:

- Invoice Number
 - Institution Name
 - Equipment Details
 - Booking ID
 - Usage Duration
 - Unit Price
 - Taxes
 - Discounts
 - Security Deposit
 - Total Amount
 - Payment Status
 - Invoice Date
 - Due Date
-

28.8 Payment Status

Status	Description
Pending	Invoice generated
Paid	Payment received
Partially Paid	Partial payment received
Failed	Payment unsuccessful
Refunded	Payment refunded
Cancelled	Invoice cancelled

28.9 Audit Logging

Every critical operation generates an audit log.

Examples:

- User Login
 - User Logout
 - User Creation
 - Equipment Added
 - Equipment Deleted
 - Booking Approved
 - Booking Rejected
 - Maintenance Updated
 - Invoice Generated
 - Payment Recorded
 - Permission Changed
-

28.10 Screens

RP-01 – Report Dashboard

Displays:

- Available Reports
 - Recent Reports
 - Scheduled Reports
 - Favorites
-

RP-02 – Report Generator

Fields:

- Report Type
- Institution
- Department
- Laboratory
- Equipment

- Date Range
- Output Format

Buttons:

- Preview
 - Generate
 - Export
-

RP-03 – Invoice Dashboard

Displays:

- Pending Invoices
 - Paid Invoices
 - Overdue Invoices
 - Total Revenue
-

RP-04 – Invoice Details

Displays:

- Invoice Information
 - Usage Details
 - Payment History
 - Download PDF
-

RP-05 – Audit Log Viewer

Displays:

- Timestamp
- User
- Action
- Module
- IP Address
- Result
- Details

Supports:

- Search

- Filter
 - Export
-

28.11 Scheduled Reports

Administrators can schedule reports:

- Daily
- Weekly
- Monthly
- Quarterly
- Yearly

Delivery Channels:

- Email
 - In-App Notification
 - Download Center
-

28.12 Database Tables

reports

Column	Type
id	BIGINT
report_name	VARCHAR
report_type	VARCHAR
generated_by	BIGINT
generated_at	TIMESTAMP
file_path	VARCHAR

invoices

Column	Type
id	BIGINT
invoice_number	VARCHAR
institution_id	BIGINT
booking_id	BIGINT
total_amount	DECIMAL
payment_status	VARCHAR
due_date	DATE
generated_at	TIMESTAMP

payments

Column	Type
id	BIGINT
invoice_id	BIGINT
payment_reference	VARCHAR
amount_paid	DECIMAL
payment_method	VARCHAR
payment_date	TIMESTAMP
payment_status	VARCHAR

audit_logs

Column	Type
id	BIGINT
user_id	BIGINT
module	VARCHAR
action	VARCHAR

entity_type	VARCHAR
entity_id	BIGINT
ip_address	VARCHAR
action_time	TIMESTAMP
result	VARCHAR

28.13 REST APIs

Generate Report

POST /api/reports/generate

Download Report

GET /api/reports/{id}/download

List Reports

GET /api/reports

Generate Invoice

POST /api/invoices

Get Invoice

GET /api/invoices/{id}

Record Payment

POST /api/payments

Get Audit Logs

GET /api/audit-logs

Export Audit Logs

GET /api/audit-logs/export

28.14 Validation Rules

- Report type is mandatory.
 - Date range must be valid.
 - Invoice number must be unique.
 - Payment amount cannot exceed outstanding balance.
 - Due date must not precede invoice date.
 - Only authorized roles can access audit logs.
 - Export format must be supported.
-

28.15 Business Rules

BR-081

Every completed shared booking generates an invoice if billing is enabled.

BR-082

Invoice numbers are generated automatically and must be unique.

BR-083

Audit logs cannot be modified or deleted by application users.

BR-084

Scheduled reports run according to the configured schedule.

BR-085

Financial reports include only finalized transactions.

BR-086

Payment status updates automatically after successful payment confirmation.

BR-087

Audit logs record both successful and failed security-sensitive operations.

BR-088

Reports respect role-based data access restrictions.

BR-089

Invoice totals include taxes, discounts, and additional charges as configured.

BR-090

Generated reports are retained according to the organization's data retention policy.

28.16 Exception Handling

Exception	HTTP Status
Report Not Found	404
Invalid Report Parameters	400
Invoice Not Found	404
Payment Failed	500
Unauthorized	401
Forbidden	403
Export Failed	500

28.17 Acceptance Criteria

ID	Requirement
AC-091	Reports generate successfully in supported formats
AC-092	Scheduled reports execute automatically
AC-093	Invoices are generated accurately
AC-094	Payments are recorded correctly
AC-095	Financial dashboards reflect current data
AC-096	Audit logs capture all configured events
AC-097	Audit logs are searchable and exportable
AC-098	Reports enforce role-based visibility
AC-099	Invoice calculations match configured pricing rules
AC-100	Compliance reports are available for audit purposes

28.18 Financial Dashboard

Displays:

- Total Revenue
 - Revenue by Institution
 - Outstanding Payments
 - Paid Invoices
 - Average Invoice Value
 - Monthly Revenue Trend
 - Top Revenue-Generating Equipment
 - Security Deposit Balance
 - Refund Summary
 - Collection Rate
-

28.19 Compliance Dashboard

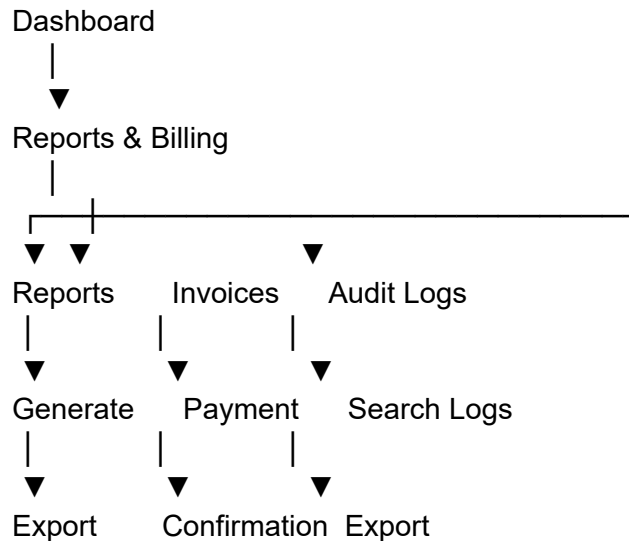
Displays:

- Total Audit Events
 - Failed Login Attempts
 - Permission Changes
 - Configuration Changes
 - Sensitive Data Access
 - Export Activities
 - Security Alerts
 - Compliance Status
-

28.20 Future Enhancements

- Integration with payment gateways (Stripe, Razorpay, PayPal)
- GST/VAT automation
- ERP integration (SAP, Oracle ERP)
- Digital invoice signing
- AI-powered financial forecasting
- Automated reconciliation
- Fraud detection analytics
- Immutable blockchain audit ledger
- Electronic compliance submissions
- Executive financial reporting portal

28.21 Screen Flow



✓ End of Part 11

At this point, the PRD covers all major functional modules:

- Authentication & Authorization
- Institution Management
- Equipment Inventory
- Booking & Scheduling
- Real-Time Analytics
- Maintenance & Calibration
- Inter-Institution Resource Sharing
- Notification & Communication
- **Reporting, Billing & Audit Management**

Next (Part 12 – Final Enterprise Sections)

The final part will include the cross-cutting sections expected in a production-ready PRD:

- Non-Functional Requirements (Performance, Security, Scalability, Availability)
- System Architecture Overview
- Integration Requirements
- Deployment Strategy
- Backup & Disaster Recovery
- Logging & Monitoring
- Security & Compliance
- Testing Strategy

- Risks & Mitigation
- Assumptions & Dependencies
- Release Plan & Milestones
- Appendix, Glossary, and Traceability Matrix

This will complete the PRD as a document suitable for handing over to architects, developers, QA engineers, and project stakeholders.

29. Non-Functional Requirements (NFR)

29.1 Overview

Non-functional requirements define **how** the system should perform rather than **what** it should do. These requirements ensure the platform is secure, scalable, reliable, maintainable, and performant.

29.2 Performance Requirements

Response Time

Operation	Maximum Response Time
Login	≤ 2 seconds
Equipment Search	≤ 2 seconds
Booking Creation	≤ 3 seconds
Dashboard Loading	≤ 4 seconds
Report Generation	≤ 30 seconds
QR Code Generation	≤ 5 seconds

Concurrent Users

The platform shall support:

- 1,000 concurrent users (initial deployment)
- 10,000 registered users
- 100 simultaneous booking requests
- 500 concurrent API requests

Database Performance

- Indexed search fields
 - Query optimization
 - Connection pooling
 - Pagination for large datasets
-

29.3 Scalability

The system shall support horizontal scaling.

Architecture should support:

- Multiple Application Servers
 - Load Balancer
 - Distributed Cache
 - Cloud Storage
 - Read Replicas
 - Container Deployment
-

29.4 Availability

Requirement	Target
System Availability	99.9%
Scheduled Maintenance	Off-Peak Hours
Maximum Downtime	< 8 hours/year
Database Recovery	< 30 minutes

29.5 Reliability

The platform shall:

- Recover from server failures automatically.

- Prevent data loss.
 - Support transactional consistency.
 - Detect service failures.
-

29.6 Security Requirements

Authentication

- JWT Authentication
- OAuth2 Login
- Password Encryption (BCrypt)

Authorization

- Role-Based Access Control (RBAC)
- Resource-Level Permissions
- API Authorization

Data Security

- HTTPS Everywhere
- TLS 1.3
- Password Hashing
- Secure Cookies
- CSRF Protection
- XSS Protection
- SQL Injection Prevention

Audit

- Login Logs
 - Activity Logs
 - Permission Changes
 - Failed Login Attempts
-

29.7 Privacy

The system shall:

- Store only necessary personal information.
- Encrypt sensitive information at rest.
- Encrypt all communication in transit.

- Allow configurable data retention policies.
 - Support anonymization for archived analytics where required.
-

29.8 Maintainability

The project shall follow:

- Clean Architecture
 - SOLID Principles
 - Layered Architecture
 - Dependency Injection
 - Modular Development
 - Code Documentation
-

29.9 Usability

The UI should be:

- Responsive
 - Mobile Friendly
 - Accessible
 - Easy Navigation
 - Consistent Design
 - Minimal Clicks
-

29.10 Compatibility

Supported Browsers

- Chrome
- Edge
- Firefox
- Safari

Supported Devices

- Desktop
- Tablet

- Mobile Browser
-

29.11 Logging

The application shall log:

- API Requests
 - Errors
 - Exceptions
 - User Activities
 - Authentication Events
 - Database Errors
-

29.12 Monitoring

Monitor:

- CPU Usage
 - Memory Usage
 - Database Health
 - API Latency
 - Error Rates
 - Storage Usage
-

29.13 Backup

Backup Strategy

Database

- Daily Incremental Backup
- Weekly Full Backup
- Monthly Archive

Files

- Daily Backup
- Versioning Enabled

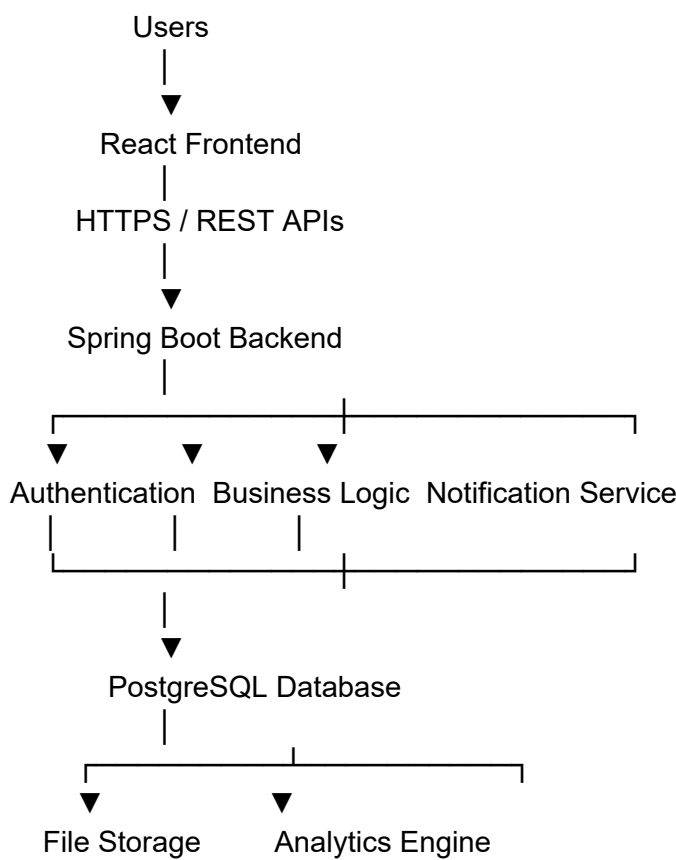
29.14 Disaster Recovery

Recovery Objectives

Metric	Target
RPO (Recovery Point Objective)	15 minutes
RTO (Recovery Time Objective)	30 minutes

30. System Architecture

30.1 High-Level Architecture



30.2 Technology Stack

Frontend

- React.js
 - React Router
 - Axios
 - Material UI (or Bootstrap)
 - Chart.js
-

Backend

- Spring Boot
 - Spring Security
 - Spring Data JPA
 - Hibernate
 - JWT
 - OAuth2
-

Database

- PostgreSQL
-

Storage

- Local Storage (Development)
 - AWS S3 / Azure Blob Storage (Production)
-

Build Tools

- Maven
 - npm
-

Version Control

- Git
- GitHub

31. Integration Requirements

External integrations may include:

- Email Service (SMTP)
 - SMS Gateway
 - QR Code Generator
 - Payment Gateway
 - Cloud Storage
 - Identity Provider (OAuth2)
 - ERP Systems (Future)
 - Calendar Services (Future)
-

32. Deployment Strategy

Development Environment

- Local Machine
 - Docker (Optional)
-

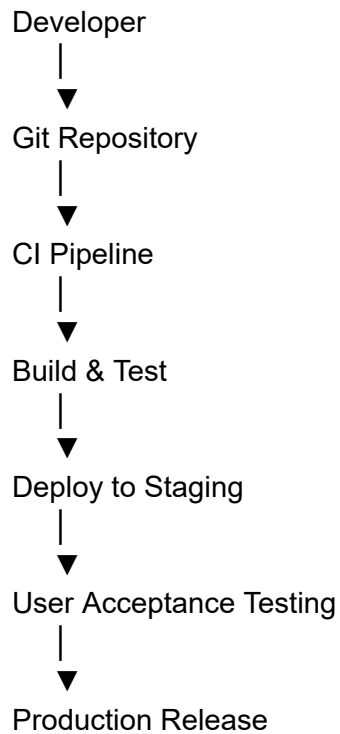
Testing Environment

- QA Server
 - Staging Database
-

Production Environment

- Linux Server
 - Nginx Reverse Proxy
 - Spring Boot Service
 - PostgreSQL Database
-

Deployment Flow



33. Testing Strategy

Test Levels

- Unit Testing
 - Integration Testing
 - System Testing
 - Regression Testing
 - User Acceptance Testing (UAT)
 - Performance Testing
 - Security Testing
-

Test Coverage

Target:

- Backend Unit Tests $\geq 80\%$
 - Critical Business Logic $\geq 90\%$
-

34. Risk Analysis

Risk	Impact	Mitigation
Server Failure	High	Redundant deployment
Database Failure	High	Replication & backups
Security Breach	High	Encryption, RBAC, monitoring
High Traffic	Medium	Load balancing
Third-party Service Failure	Medium	Retry mechanisms & graceful degradation
Data Corruption	High	Transaction management & backups

35. Assumptions

- All institutions have reliable internet access.
 - Equipment data is maintained accurately.
 - Users receive required training.
 - Authentication services are available.
 - Partner institutions comply with sharing agreements.
-

36. Dependencies

- React Frontend
 - Spring Boot Backend
 - PostgreSQL Database
 - Email Service
 - File Storage
 - Notification Service
 - QR Code Library
 - Payment Gateway (Optional)
-

37. Release Plan

Phase 1

- Authentication
 - Institution Management
 - Equipment Inventory
-

Phase 2

- Booking
 - Maintenance
 - Notifications
-

Phase 3

- Analytics
 - Resource Sharing
 - Billing
-

Phase 4

- AI Features
 - Mobile App
 - ERP Integrations
-

38. Success Metrics

The project will be considered successful if it achieves:

KPI	Target
Equipment Utilization	> 80%
Booking Success Rate	> 95%
Average Booking Response Time	< 3 seconds

User Satisfaction Score	> 4.5 / 5
System Availability	≥ 99.9%
Preventive Maintenance Compliance	> 95%
Report Generation Success	> 99%
Notification Delivery Success	> 98%

39. Glossary

Term	Meaning
Equipment	Laboratory asset managed by the system
Booking	Reservation of equipment for a defined time slot
Calibration	Verification of measurement accuracy
Work Order	Maintenance task assigned to a technician
Institution	University or research organization using the platform
Department	Academic or research division within an institution
Laboratory	Physical location where equipment is housed
Utilization	Percentage of available time during which equipment is actively used
RBAC	Role-Based Access Control
KPI	Key Performance Indicator

40. Requirements Traceability Matrix (RTM)

Business Goal	Module	Requirement IDs
Secure access	Authentication	AC-001 to AC-010

Organizational management	Institution Management	AC-021 to AC-030
Asset management	Equipment Inventory	AC-031 to AC-040
Equipment scheduling	Booking	AC-041 to AC-050
Operational insights	Analytics	AC-051 to AC-060
Equipment reliability	Maintenance	AC-061 to AC-070
Collaboration	Resource Sharing	AC-071 to AC-080
User communication	Notifications	AC-081 to AC-090
Governance & compliance	Reporting/Billing/Audit	AC-091 to AC-100

41. Appendix

The appendix may include:

- Wireframes
 - UI mockups
 - ER diagrams
 - API specifications (OpenAPI/Swagger)
 - Database schema scripts
 - Sample reports
 - User manuals
 - Deployment guides
 - Configuration guides
 - Test cases
-

42. Document Sign-Off

Role	Responsibility
Product Manager	Approves business requirements
Solution Architect	Validates technical feasibility
UI/UX Designer	Confirms design alignment
Development Lead	Reviews implementation readiness

QA Lead	Reviews testability
Project Manager	Approves project scope
Client/Stakeholder	Final business approval



End of Enterprise PRD

This completes a comprehensive enterprise-style Product Requirements Document. It includes:

- **42 structured sections**
- **9 major functional modules**
- **100 acceptance criteria**
- **90 business rules**
- **REST API definitions**
- **Database schema**
- **Screen inventories**
- **Workflows and process flows**
- **Architecture and deployment guidance**
- **Testing, security, reporting, governance, and operational requirements**

This document is suitable as a foundation for design, development, QA planning, and project execution, while leaving room for detailed UI designs, OpenAPI specifications, and low-level technical design documents that are typically produced in subsequent project phases.